
Convergence Analysis of the Lion Optimizer in Centralized and Distributed Settings

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Abstract

In this paper, we provide a comprehensive convergence analysis for the Lion optimizer. First, we establish that the original Lion achieves a convergence rate of $\mathcal{O}(d^{1/2}T^{-1/4})$, where d denotes the problem dimension and T is the iteration number. To improve this rate, we propose a variance reduction variant of Lion, which attains an enhanced rate of $\mathcal{O}(d^{1/2}T^{-1/3})$ with the average smoothness assumption. Then, we extend our analysis to distributed settings. We demonstrate that the distributed Lion optimizer and its variance reduction counterpart achieve linear speedup with respect to the number of nodes n , yielding convergence rates of $\mathcal{O}(d^{1/2}(nT)^{-1/4})$ and $\mathcal{O}(d^{1/2}(nT)^{-1/3})$, respectively. Additionally, we investigate a communication-efficient distributed Lion variant that utilizes sign compression for bidirectional communication. By employing unbiased sign operations, this variant achieves a convergence rate of $\mathcal{O}\left(\max\left\{\frac{d^{1/4}}{T^{1/4}}, \frac{d^{1/10}}{n^{1/5}T^{1/5}}\right\}\right)$, and its variance-reduced counterpart can further improve the rate to $\mathcal{O}\left(\frac{d^{1/4}}{T^{1/4}}\right)$. Finally, we conduct numerical experiments to validate the effectiveness of the proposed methods.

1. Introduction

The Lion (evolved sign momentum) optimizer (Chen et al., 2023), introduced by Google through program search, is a highly efficient optimization algorithm known for its memory efficiency and strong generalization performance. It has demonstrated effectiveness across a broad range of tasks,

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including fine-tuning large language models, training diffusion models, and vision-language contrastive learning, among others. In this paper, we investigate the convergence properties of the Lion optimizer within the framework of stochastic optimization, which is formulated as:

$$\min_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x}), \quad (1)$$

where $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is a smooth and non-convex function. We assume that only noisy unbiased gradient estimations are available, denoted as $\nabla f(\mathbf{x}; \xi)$, where ξ represents a random sample drawn from a stochastic oracle, such that $\mathbb{E}[\nabla f(\mathbf{x}; \xi)] = \nabla f(\mathbf{x})$.

Although the Lion optimizer exhibits superior empirical performance, its theoretical properties have not been fully explored. Previous works (Dong et al., 2024) have derived a convergence rate of $\mathcal{O}(d^{1/2}T^{-1/4})$, but this result relies on the additional assumption of function coerciveness. A similar result was obtained by Sfyraki & Wang (2025), although their analysis depends on large batch sizes and the theoretical result is derived in terms of the Frank-Wolfe gap for constrained problems. In contrast, this paper focuses on the convergence to the stationary point for the unconstrained problem (1), in which we establish the convergence rate of $\mathcal{O}(d^{1/2}T^{-1/4})$ under standard assumptions. Furthermore, by incorporating variance reduction estimators, we show that the convergence rate can be further improved to $\mathcal{O}(d^{1/2}T^{-1/3})$ with a slight modification.

The study of the Lion optimizer extends beyond centralized settings. However, existing literature on distributed optimization often requires strong additional assumptions to derive convergence guarantees. Notably, Liu et al. (2024) analyze the convergence properties for constrained problems under the assumption of bias correction, which assumes that the expectation of the stochastic gradient and its sign share the same sign. Additionally, Tang & Chang (2024) provide convergence guarantees for the FedLion algorithm but impose the strong assumption of bounded heterogeneity. In contrast, we aim to provide theoretical guarantees for the distributed Lion optimizer under standard assumptions. Specifically, we establish a convergence rate of $\mathcal{O}(d^{1/2}(nT)^{-1/4})$ for the distributed Lion, where n denotes the number of nodes in the distributed system.

Table 1. Summary of convergence rates for the Lion-based algorithms under centralized and distributed settings.

Settings	Analysis	Convergence Rates	Additional Requirements
Centralized	Dong et al. (2024)	$\mathcal{O}\left(\frac{d^{1/2}}{T^{1/4}}\right)$	coerciveness
	Theorem 3.6	$\mathcal{O}\left(\frac{d^{1/2}}{T^{1/4}}\right)$	–
	Theorem 3.7	$\mathcal{O}\left(\frac{d^{1/2}}{T^{1/3}}\right)$	average smoothness
Distributed	Theorem 4.5	$\mathcal{O}\left(\frac{d^{1/2}}{(nT)^{1/4}}\right)$	–
	Theorem 4.6	$\mathcal{O}\left(\frac{d^{1/2}}{(nT)^{1/3}}\right)$	average smoothness
Distributed with 1-bit compression	Theorem 5.3	$\mathcal{O}\left(\frac{d^{1/2}}{T^{1/2}} + \frac{d}{n^{1/2}}\right)$	–
		$\mathcal{O}\left(\frac{n^{1/2}}{T} + \frac{d}{n^{1/2}}\right)$	–
	Theorem 5.4	$\mathcal{O}\left(\max\left\{\frac{d^{1/4}}{T^{1/4}}, \frac{d^{1/10}}{n^{1/5}T^{1/5}}\right\}\right)$	–
	Theorem 5.5	$\mathcal{O}\left(\frac{d^{1/4}}{T^{1/4}}\right)$	average smoothness

By incorporating variance reduction estimators, we further improve the rate to $\mathcal{O}(d^{1/2}(nT)^{-1/3})$, matching the convergence rates obtained in centralized settings.

Furthermore, we explore the communication-efficient variants of the distributed Lion optimizer. While the original Lion optimizer utilizes sign operations for updates, its distributed version can only ensure 1-bit sign compression in a single communication direction. By employing the sign operator in both the server and client nodes, we can ensure 1-bit sign compression in both directions. For this new method, we derive convergence rates of $\mathcal{O}\left(\frac{d^{1/2}}{T^{1/2}} + \frac{d}{n^{1/2}}\right)$ and $\mathcal{O}\left(\frac{n^{1/2}}{T} + \frac{d}{n^{1/2}}\right)$. Moreover, by using unbiased sign operations in both communication directions, we further improve the convergence rates to $\mathcal{O}\left(\max\left\{\frac{d^{1/4}}{T^{1/4}}, \frac{d^{1/10}}{n^{1/5}T^{1/5}}\right\}\right)$ and $\mathcal{O}\left(\frac{d^{1/4}}{T^{1/4}}\right)$ for the distributed Lion variant and its variance reduction counterpart. In summary, the contributions of this paper are as follows:

- We establish a convergence rate of $\mathcal{O}(d^{1/2}T^{-1/4})$ for the Lion optimizer under standard assumptions, without relying on the coerciveness assumption used in previous work (Dong et al., 2024). Additionally, we improve this rate to $\mathcal{O}(d^{1/2}T^{-1/3})$ by employing variance reduction techniques.
- In distributed settings, we show that the distributed Lion optimizer and its variance reduction version achieve convergence rates of $\mathcal{O}(d^{1/2}(nT)^{-1/4})$ and $\mathcal{O}(d^{1/2}(nT)^{-1/3})$, respectively, achieving the effect of linear speedup.
- Finally, we introduce communication-efficient variants of the distributed Lion optimizer, which ensure 1-bit sign compression in both communication direc-

tions. These variants can achieve convergence rates of $\mathcal{O}\left(\frac{d^{1/2}}{T^{1/2}} + \frac{d}{n^{1/2}}\right)$ and $\mathcal{O}\left(\frac{n^{1/2}}{T} + \frac{d}{n^{1/2}}\right)$. By employing unbiased sign operations in both communication directions, we further improve the convergence rates to $\mathcal{O}\left(\max\left\{\frac{d^{1/4}}{T^{1/4}}, \frac{d^{1/10}}{n^{1/5}T^{1/5}}\right\}\right)$ and $\mathcal{O}\left(\frac{d^{1/4}}{T^{1/4}}\right)$.

We compare our results with existing methods in Table 1.

2. Related work

This section reviews related work on the theoretical analysis of the Lion optimizer, stochastic optimization and variance reduction, and sign-based methods.

2.1. Theoretical analysis of the Lion optimizer

The Lion optimizer (Chen et al., 2023), originally discovered by Google through symbolic program search, has shown superior performance in training large AI models, including large language models. It often outperforms AdamW in terms of both training speed and memory efficiency. Despite its empirical success, the theoretical foundations of Lion were not fully understood until recent studies. A notable contribution by Chen et al. (2024) demonstrates that Lion operates as a constrained optimization algorithm, effectively minimizing a general loss function $f(\mathbf{x})$ subject to the constraint $\|\mathbf{x}\|_\infty \leq 1/\lambda$. This is achieved through the incorporation of decoupled weight decay, where λ represents the weight decay coefficient.

Further research (Dong et al., 2024) establishes that the Lion optimizer converges to Karush-Kuhn-Tucker (KKT) points at a rate of $\mathcal{O}(d^{1/2}T^{-1/4})$ for constrained optimization problems, where d represents the problem dimension. This

convergence result extends to the unconstrained case with the same rate, under the additional assumption of coerciveness of the objective function. Subsequent work by Sfyraiki & Wang (2025) highlights that Lion can be viewed as a special case of the Stochastic Frank-Wolfe algorithm, establishing a similar convergence guarantee in terms of the Frank-Wolfe gap by using large batch sizes.

In the context of distributed optimization, Ishikawa et al. (2024) integrate quantization techniques into the distributed Lion, although they do not provide theoretical guarantees for their approach. While some literature analyzes the convergence properties of distributed variants of Lion, these studies typically rely on additional assumptions. For instance, Liu et al. (2024) derive convergence guarantees for constrained problems with the assumption of bias correction, which requires the expected values of the stochastic gradient and its sign to align. Additionally, Tang & Chang (2024) analyze the FedLion algorithm under the strong assumption of bounded heterogeneity.

2.2. Stochastic optimization and variance reduction

Stochastic optimization has been widely investigated in the machine learning community, focusing on studying the convergence properties of algorithms in stochastic settings. It is well-known that the traditional stochastic gradient descent (SGD) method can achieve a convergence rate of $\mathcal{O}(T^{-1/4})$ (Ghadimi & Lan, 2013), which matches the lower bound under the standard smoothness assumption (Arjevani et al., 2023). By incorporating past gradients into the current update, the momentum technique is introduced to accelerate SGD, which helps escape saddle points faster (Wang et al., 2020), while maintaining the same convergence rate (Liu et al., 2020).

To further improve the convergence rate, variance reduction techniques have been developed to accelerate convergence by reducing the variance in gradient estimates. Classical algorithms include SAG (Roux et al., 2012), SVRG (Zhang et al., 2013; Johnson & Zhang, 2013), and SARAH (Nguyen et al., 2017), which achieve improved convergence rates for convex or strongly convex functions. For non-convex functions, the SPIDER algorithm (Fang et al., 2018) improves the convergence of SGD to $\mathcal{O}(T^{-1/3})$ under the average smoothness assumption. This rate can be further improved to $\mathcal{O}(n^{1/4}T^{-1/2})$ for finite-sum problems, where n is the number of functions in the finite-sum structure. To avoid the large batch sizes required in the SPIDER algorithm, Cutkosky & Orabona (2019) propose the STORM method, which attains the same $\mathcal{O}(T^{-1/3})$ convergence rate with constant batch sizes. Recent studies also focus on more complicated stochastic optimization problems such as two-level optimization (Hu et al., 2019; Yang et al., 2021; Jiang et al., 2022a; Zhang et al., 2023; Jiang et al., 2026) and

multi-level problems (Balasubramanian et al., 2021; Xiao et al., 2022; Jiang et al., 2022b; 2024a; Zhang et al., 2024; Jiang et al., 2025b; Zhang & Gao, 2025), and provide convergence analysis for them.

2.3. Sign-based optimization methods

Sign-based optimization methods, such as SignSGD, have gained popularity due to their simplicity and communication efficiency, especially in distributed training scenarios. These methods update model parameters using only the sign of the gradient, thereby reducing the amount of data transmitted between nodes. The convergence of SignSGD methods was first studied by Bernstein et al. (2018; 2019), establishing a convergence rate of $\mathcal{O}(d^{1/2}T^{-1/4})$, where d is the problem dimension. To avoid the large batch sizes or the unimodal symmetric noise assumption used in earlier analyses (Bernstein et al., 2018; 2019), later studies (Sun et al., 2023; Jiang et al., 2025c) introduce the momentum technique, achieving a similar convergence rate for SignSGD under more standard assumptions. This rate was further improved to $\mathcal{O}(d^{1/2}T^{-1/3})$ by Jiang et al. (2024b), using variance reduction techniques.

In distributed settings, sign-based methods with majority voting have also been widely investigated due to their communication efficiency. In homogeneous environments, Bernstein et al. (2018; 2019) demonstrate that SignSGD with majority voting achieves a convergence rate of $\mathcal{O}(d^{1/2}T^{-1/4})$. For more challenging heterogeneous environments, the MV-sto-signSGD-SIM method (Sun et al., 2023) achieves a convergence rate of $\mathcal{O}(dT^{-1/4} + dn^{-1/2})$, where n represents the number of nodes in the distributed system. This rate is further improved to $\mathcal{O}(d^{1/2}T^{-1/2} + dn^{-1/2})$ by Jiang et al. (2025c). To ensure that the gradient norm continues to decrease as T increases, Jin et al. (2021) prove that Stochastic-Sign SGD can achieve a convergence rate of $\mathcal{O}(d^{3/8}T^{-1/8})$ in terms of the l_2 -norm, which is further improved to $\mathcal{O}(d^{1/4}T^{-1/4})$ using the variance reduction estimators (Jiang et al., 2024b). More recent studies focus on different settings, such as l_∞ -smoothness (Tao et al., 2026), non-smooth gradient objectives (Yu et al., 2026a), heavy-tailed noise (Yu et al., 2026b), and second-order gradient Lipschitz (Sun & Liu, 2026).

3. Convergence analysis for the Lion optimizer

In this section, we first outline the assumptions used in our analysis throughout this section, followed by the convergence guarantees for the traditional Lion optimizer. To further accelerate the convergence rate, we introduce a variance reduction version of the Lion optimizer and present an improved convergence result.

3.1. Assumptions

We begin by listing the assumptions required for our analysis, which are standard in the study of convergence rates in stochastic optimization.

Assumption 3.1. (Smoothness) The objective function f is L -smooth, i.e., $\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\| \leq L \|\mathbf{x} - \mathbf{y}\|$.

For the analysis of the Lion optimizer with variance reduction, we introduce the following average smoothness assumption, which is also commonly used in the variance reduction literature (Nguyen et al., 2017; Fang et al., 2018; Wang et al., 2019; Cutkosky & Orabona, 2019; Levy et al., 2021; Li et al., 2021).

Assumption 3.2. (Average smoothness) The stochastic objective function $f(\cdot; \xi)$ is L -smooth, satisfying

$$\mathbb{E}_\xi \left[\|\nabla f(\mathbf{x}; \xi) - \nabla f(\mathbf{y}; \xi)\|^2 \right] \leq L^2 \|\mathbf{x} - \mathbf{y}\|^2.$$

Assumption 3.3. The stochastic gradient is unbiased and its noise is bounded.

$$\begin{aligned} \mathbb{E}_\xi [\nabla f(\mathbf{x}; \xi)] &= \nabla f(\mathbf{x}), \\ \mathbb{E}_\xi \left[\|\nabla f(\mathbf{x}; \xi) - \nabla f(\mathbf{x})\|^2 \right] &\leq \sigma^2. \end{aligned}$$

Assumption 3.4. $f_* = \inf_{\mathbf{x}} f(\mathbf{x}) \geq -\infty$ and $f(\mathbf{x}_1) - f_* \leq \Delta_f$ for the initial solution $\mathbf{x}_1$.

Remark: Previous analysis of the Lion optimizer assumes that the objective function f is coercive, i.e.,

$$\lim_{\|\mathbf{x}\| \rightarrow \infty} f(\mathbf{x}) \rightarrow \infty,$$

which is not required in our analysis.

3.2. Convergence rate for the traditional Lion optimizer

We first introduce the original Lion optimizer in Algorithm 1 (v1). The Lion optimizer uses momentum estimators to track the gradient, denoted as $\mathbf{v}_t$ and $\mathbf{m}_t$, and utilizes the sign operation of $\mathbf{v}_t$ to compute the updates. The decoupled weight decay is also incorporated, with λ representing the strength of the decay. In the first time step ($t = 1$), we initialize as $\mathbf{v}_1 = \mathbf{m}_1 = \nabla f(\mathbf{x}_1; \xi_1)$. For the update used in the algorithm, we first present the following lemma.

Lemma 3.5. For the decision variable update $\mathbf{x}_{t+1} = \mathbf{x}_t - \eta (\text{sign}(\mathbf{v}_t) + \lambda \mathbf{x}_t)$, we can ensure that

$$\frac{1}{T} \sum_{t=1}^T \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 \leq 6\eta^2 d + \frac{2\eta\lambda \|\mathbf{x}_1\|^2}{T}.$$

Remark: Using the above lemma in our theoretical analysis allows us to avoid the coerciveness assumption, which is

typically required in previous literature. This improvement is due to the more careful analysis in the above lemma.

Next, we provide the guarantee for the Lion optimizer.

Theorem 3.6. Under Assumptions 3.1, 3.3 and 3.4, setting $\beta_2^2 \leq \beta_1 \leq \sqrt{\beta_2}$, $\beta_2 = \mathcal{O}(T^{-1/2})$, $\eta = \mathcal{O}(d^{-1/2}T^{-3/4})$, $\lambda = \mathcal{O}(d^{1/2}T^{-1/4})$, and $\|\mathbf{x}_1\|_\infty \leq \mathcal{O}(1/\lambda)$, the Lion optimizer (v1) ensures that

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E} [\|\nabla f(\mathbf{x}_t)\|_1] \leq \mathcal{O} \left(\frac{(1 + \Delta_f + \sigma + L) d^{1/2}}{T^{1/4}} \right).$$

Remark: The above result matches the lower bound of $\mathcal{O}(T^{-1/4})$ for stochastic optimization under the standard smoothness assumption (Arjevani et al., 2023). Compared to previous work, we achieve this rate without the need for additional assumptions, such as the coerciveness of the objective function (Dong et al., 2024).

Dimensional dependence: The $d^{1/2}$ factor arises from the ℓ_1 -norm analysis and is standard for the ℓ_1 measure. Existing convergence analyses under the ℓ_1 -norm criterion and standard ℓ_2 -smoothness assumptions also exhibit a $d^{1/2}$ dependence (Liu et al., 2025; Li et al., 2025a;b). Although the lower bound for ℓ_1 -norm is not yet established, both empirical and theoretical evidence suggest that this $d^{1/2}$ dependence is likely unavoidable.

- *Empirical evidence.* Prior works (Bernstein et al., 2018; Li et al., 2025a;b) conducted extensive experiments on vision and language tasks and observed that the ratio $r = \|\nabla f(x)\|_1 / \|\nabla f(x)\|_2$ consistently remains at the scale $\Theta(\sqrt{d})$. This supports the appearance of the $d^{1/2}$ factor with the ℓ_1 norm.
- *Theoretical evidence.* Jiang et al. (2025a) proves the first $\Omega \left(\sqrt{\frac{d\|L\|_\infty}{T}} + \frac{d^{1/4}(\sum_{i=1}^d \sigma_i \sqrt{L_i})^{1/2}}{T^{1/4}} \right)$ lower bound for SGD under the ℓ_1 -norm measure. If σ_i and L_i are uniform across coordinates, such as $\sigma_i = \sigma/\sqrt{d}$, $L_i = L$, then the bound simplifies to $\Omega \left(\frac{\sqrt{d}}{T^{1/4}} \right)$, which confirms that the $d^{1/2}$ factor is inherent in the ℓ_1 -norm setting.

3.3. Lion optimizer with variance reduction

While the $\mathcal{O}(T^{-1/4})$ convergence rate is optimal under standard smoothness assumption, it is well-known that this rate can be further improved using the average smoothness condition (Fang et al., 2018; Wang et al., 2019; Cutkosky & Orabona, 2019). In this subsection, we introduce the Lion optimizer with variance reduction (Lion-VR) to achieve a faster convergence rate.

Note that the traditional Lion optimizer uses two different momentum-based estimators $\mathbf{v}_t$ and $\mathbf{m}_t$ to track the gradient.

Algorithm 1 Lion (v1: original version, v2: variance reduction version.)

- 1: **Input:** Momentum parameters β_1, β_2, γ , learning rate η , weight decay parameter λ .
 - 2: **for** time step $t = 1$ **to** T **do**
 - 3: Update $\mathbf{v}_t = (1 - \beta_1)\mathbf{m}_{t-1} + \beta_1 \nabla f(\mathbf{x}_t; \xi_t)$
 - 4: (v1) Update $\mathbf{m}_t = (1 - \beta_2)\mathbf{m}_{t-1} + \beta_2 \nabla f(\mathbf{x}_t; \xi_t)$
 - 5: (v2) Update $\mathbf{m}_t = (1 - \beta_2)\mathbf{m}_{t-1} + \beta_2 \nabla f(\mathbf{x}_t; \xi_t) + \gamma (\nabla f(\mathbf{x}_t; \xi_t) - \nabla f(\mathbf{x}_{t-1}; \xi_t))$
 - 6: Update $\mathbf{x}_{t+1} = \mathbf{x}_t - \eta (\text{sign}(\mathbf{v}_t) + \lambda \mathbf{x}_t)$
 - 7: **end for**
-

For our Lion-VR method, we modify the $\mathbf{m}_t$ estimator using the variance reduction technique STORM (Cutkosky & Orabona, 2019). Specifically, for the first time step ($t = 1$), we simply initialize as $\mathbf{v}_1 = \mathbf{m}_1 = \frac{1}{B_0} \sum_{i=1}^{B_0} \nabla f(\mathbf{x}_1; \xi_1^i)$, where $B_0 = \mathcal{O}(T^{1/3})$ is the batch size only used in the initial iteration. For other iterations, we use

$$\mathbf{m}_t = (1 - \beta_2)\mathbf{m}_{t-1} + \beta_2 \nabla f(\mathbf{x}_t; \xi_t) + \gamma (\nabla f(\mathbf{x}_t; \xi_t) - \nabla f(\mathbf{x}_{t-1}; \xi_t)).$$

The difference lies in the last term, which ensures that the estimation error $\mathbb{E} [\|\nabla f(\mathbf{x}_t) - \mathbf{v}_t\|^2]$ decays more rapidly over time. This newly proposed algorithm is outlined in Algorithm 1 (v2), named Lion-VR, with the modification appearing in step 5. We now establish the improved convergence guarantee for this method in the following theorem.

Theorem 3.7. *Under Assumptions 3.2, 3.3 and 3.4, by setting $\beta_2 \leq \beta_1 \leq \sqrt{\beta_2}$, $\beta_2 = \mathcal{O}(T^{-2/3})$, $\gamma = 1 - \beta_2$, $\eta = \mathcal{O}(d^{-1/2}T^{-2/3})$, $B_0 = \mathcal{O}(T^{1/3})$, $\lambda = \mathcal{O}(d^{1/2}T^{-1/3})$, and that $\|\mathbf{x}_1\|_\infty \leq \mathcal{O}(1/\lambda)$, the Lion-VR optimizer ensures*

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E} [\|\nabla f(\mathbf{x}_t)\|_1] \leq \mathcal{O} \left(\frac{(1 + \Delta_f + \sigma + L) d^{1/2}}{T^{1/3}} \right).$$

Remark: The Lion-VR method achieves a better convergence rate of $\mathcal{O}(T^{-1/3})$, which matches the optimal bound for stochastic optimization under the average smoothness assumption (Li et al., 2021; Arjevani et al., 2023).

4. Convergence analysis for distributed Lion

In this section, we investigate the convergence of the Lion optimizer in distributed settings. We first introduce the problem setup and the assumptions used, followed by the distributed Lion optimizer and its variance reduction version, along with their corresponding convergence guarantees.

4.1. Problem setup and assumptions

We consider the following distributed learning problem:

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x}) &:= \frac{1}{n} \sum_{j=1}^n f_j(\mathbf{x}), \\ f_j(\mathbf{x}) &= \mathbb{E}_{\xi^j \sim \mathcal{D}_j} [f_j(\mathbf{x}; \xi^j)], \end{aligned} \quad (2)$$

where $\mathcal{D}_j$ denotes the data distribution on node j , and $f_j(\mathbf{x})$ is the corresponding loss function. Unlike the homogeneous setting, where the data distribution $\mathcal{D}_j$ and the loss function f_j are assumed to be identical across nodes, we investigate the more challenging heterogeneous setting, where the data and loss functions on each node can differ significantly.

Next, we list the assumptions used in our analysis, which are very similar to those used in the centralized settings.

Assumption 4.1. (Smoothness) The objective function on each node is L -smooth, such that

$$\|\nabla f_j(\mathbf{x}) - \nabla f_j(\mathbf{y})\| \leq L \|\mathbf{x} - \mathbf{y}\|.$$

For the variance reduction version of the Lion optimizer, we use the average smoothness assumption instead of the smoothness assumption.

Assumption 4.2. (Average smoothness) The stochastic objective function on each node is L -smooth such that

$$\mathbb{E}_\xi [\|\nabla f_j(\mathbf{x}; \xi) - \nabla f_j(\mathbf{y}; \xi)\|^2] \leq L^2 \|\mathbf{x} - \mathbf{y}\|^2.$$

Assumption 4.3. The stochastic gradient on each node is unbiased, and its noise is bounded, such that

$$\begin{aligned} \mathbb{E}_\xi [\nabla f_j(\mathbf{x}; \xi)] &= \nabla f_j(\mathbf{x}), \\ \mathbb{E}_\xi [\|\nabla f_j(\mathbf{x}; \xi) - \nabla f_j(\mathbf{x})\|^2] &\leq \sigma^2. \end{aligned}$$

Assumption 4.4. $f_* = \inf_{\mathbf{x}} f(\mathbf{x}) \geq -\infty$ and $f(\mathbf{x}_1) - f_* \leq \Delta_f$ for the initial solution $\mathbf{x}_1$.

4.2. Distributed Lion optimizers and their convergence guarantees

In the distributed setting, we first compute the momentum estimators $\mathbf{v}_t^j$ and $\mathbf{m}_t^j$ on each node j (For the first time step, i.e., $t = 1$, we initialize $\mathbf{m}_1^j = \mathbf{v}_1^j = \nabla f(\mathbf{x}_1^j; \xi_1^j)$). Then, the parameter server collects each gradient estimator $\mathbf{v}_t^j$ from the nodes and computes $\mathbf{v}_t = \sum_{j=1}^n \mathbf{v}_t^j$. This vector $\text{sign}(\mathbf{v}_t)$ is then sent back to each node, which uses it to update the model parameters (Note that $\text{sign}(\sum_{j=1}^n \mathbf{v}_t^j)$ and $\text{sign}(\frac{1}{n} \sum_{j=1}^n \mathbf{v}_t^j)$ are equivalent). The complete algorithm is outlined in Algorithm 2 (v1), referred to as the distributed Lion (Dis-Lion) optimizer. Next, we provide the theoretical guarantees for this algorithm.

Algorithm 2 Distributed Lion Optimizer

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1: Input: Momentum parameters  $\beta_1, \beta_2, \gamma_1, \gamma_2$ , learning rate  $\eta$ , weight decay parameter  $\lambda$ .
2: for time step  $t = 1$  to  $T$  do
3:   for node  $j = 1$  to  $n$  do
4:     (v1) Update  $\mathbf{v}_t^j = (1 - \beta_1)\mathbf{m}_{t-1}^j + \beta_1 \nabla f_j(\mathbf{x}_t; \xi_t^j)$ 
5:     Update  $\mathbf{m}_t^j = (1 - \beta_2)\mathbf{m}_{t-1}^j + \beta_2 \nabla f_j(\mathbf{x}_t; \xi_t^j)$ 
6:     (v2) Update  $\mathbf{v}_t^j = (1 - \beta_1)\mathbf{m}_{t-1}^j + \beta_1 \nabla f_j(\mathbf{x}_t; \xi_t^j) + \gamma_1 \left( \nabla f_j(\mathbf{x}_t; \xi_t^j) - \nabla f_j(\mathbf{x}_{t-1}; \xi_t^j) \right)$ 
7:     Update  $\mathbf{m}_t^j = (1 - \beta_2)\mathbf{m}_{t-1}^j + \beta_2 \nabla f_j(\mathbf{x}_t; \xi_t^j) + \gamma_2 \left( \nabla f_j(\mathbf{x}_t; \xi_t^j) - \nabla f_j(\mathbf{x}_{t-1}; \xi_t^j) \right)$ 
8:     Compute  $\mathbf{v}_t = \sum_{j=1}^n \mathbf{v}_t^j$  # collect and share to all nodes
9:     Update  $\mathbf{x}_{t+1} = \mathbf{x}_t - \eta (\text{sign}(\mathbf{v}_t) + \lambda \mathbf{x}_t)$ 
10:   end for
11: end for
    
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Theorem 4.5. Under Assumptions 4.1, 4.3 and 4.4, by setting that $T \geq n$, $\beta_2^2 \leq \beta_1 \leq \sqrt{\beta_2}$, $\beta_2 = \mathcal{O}(n^{1/2}T^{-1/2})$, $\eta = \mathcal{O}(n^{1/4}d^{-1/2}T^{-3/4})$, $\lambda \leq \mathcal{O}(n^{-1/4}d^{1/2}T^{-1/4})$, and $\|\mathbf{x}_1\|_\infty \leq \mathcal{O}(1/\lambda)$, distributed Lion optimizer (v1) ensures

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E} [\|\nabla f(\mathbf{x}_t)\|_1] \leq \mathcal{O} \left(\frac{(1 + \Delta_f + \sigma + L) d^{1/2}}{(nT)^{1/4}} \right).$$

To improve the convergence rate using the average smoothness assumption, we further introduce the variance reduction technique into the momentum estimators $\mathbf{m}_t$ and $\mathbf{v}_t$. Similar to the centralized setting, we apply the STORM estimator on each node j . For the first time step ($t = 1$), we simply initialize $\mathbf{v}_1^j = \mathbf{m}_1^j = \frac{1}{B_0} \sum_{i=1}^{B_0} \nabla f(\mathbf{x}_1; \xi_1^{j,i})$, where $B_0 = \mathcal{O}(n^{-2/3}T^{1/3})$ is the batch size only used in the first iteration. For other iterations, we employ

$$\begin{aligned} \mathbf{v}_t^j &= (1 - \beta_1)\mathbf{m}_{t-1}^j + \beta_1 \nabla f_j(\mathbf{x}_t; \xi_t^j) \\ &\quad + \gamma_1 \left(\nabla f_j(\mathbf{x}_t; \xi_t^j) - \nabla f_j(\mathbf{x}_{t-1}; \xi_t^j) \right), \\ \mathbf{m}_t^j &= (1 - \beta_2)\mathbf{m}_{t-1}^j + \beta_2 \nabla f_j(\mathbf{x}_t; \xi_t^j) \\ &\quad + \gamma_2 \left(\nabla f_j(\mathbf{x}_t; \xi_t^j) - \nabla f_j(\mathbf{x}_{t-1}; \xi_t^j) \right). \end{aligned}$$

The resulting algorithm is outlined in Algorithm 2 (v2), referred to as Dis-Lion-VR, with the modification appearing in steps 6 and 7. Then, we present the improved convergence rate for this method.

Theorem 4.6. Under Assumptions 4.2, 4.3 and 4.4, setting $\beta_1 \leq \sqrt{\beta_2}$, $\beta_2 = \mathcal{O}(n^{1/3}T^{-2/3})$, $\gamma_1 = 1 - \beta_1$, $\gamma_2 = 1 - \beta_2$, $\eta = \mathcal{O}(n^{1/3}d^{-1/2}T^{-2/3})$, $\lambda = \mathcal{O}(n^{-1/3}d^{1/2}T^{-1/3})$, $B_0 = \mathcal{O}(n^{-2/3}T^{1/3})$, $T \geq n^2$, and $\|\mathbf{x}_1\|_\infty \leq \mathcal{O}(1/\lambda)$, the distributed Lion optimizer (v2) ensures

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E} [\|\nabla f(\mathbf{x}_t)\|_1] \leq \mathcal{O} \left(\frac{(1 + \Delta_f + \sigma + L) d^{1/2}}{(nT)^{1/3}} \right).$$

Linear speedup Theorems 4.5 and 4.6 establish rates of $\mathcal{O}(d^{1/2}/(nT)^{1/4})$ and $\mathcal{O}(d^{1/2}/(nT)^{1/3})$, respectively.

Therefore, to achieve an ϵ -stationary point, the required iteration number T scales as $\mathcal{O}(d^2/(n\epsilon^4))$ and $\mathcal{O}(d^{3/2}/(n\epsilon^3))$, which is a factor- n reduction when n workers are used.

5. Communication-efficient distributed Lion

In the previous section, we note that the distributed Lion optimizer is communication-efficient, as the information sent back to each node is 1-bit sign information (e.g., $\text{sign}(\mathbf{v}_t)$). However, during the update, each node must send the gradient estimator $\mathbf{v}_t^j$ to the parameter server, which is no longer 1-bit sign information. As a result, a natural question arises: can each node send only the sign information of $\mathbf{v}_t^j$ to the server, thereby reducing communication cost even further? In this section, we investigate the distributed Lion optimizer with 1-bit sign compression in both directions.

The most straightforward approach is to apply the sign operation twice, leading to the following update rule:

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \eta \left(\text{sign} \left(\frac{1}{n} \sum_{j=1}^n \text{sign}(\mathbf{v}_t^j) \right) + \lambda \mathbf{x}_t \right).$$

However, the sign operation is a biased estimator, and applying it twice may introduce significant estimation error. To mitigate this, we use the following unbiased sign operation, which is widely adopted in previous sign-based optimization algorithms (Jin et al., 2021; Sun et al., 2023).

Definition 5.1. For any vector $\mathbf{v}$ with $\|\mathbf{v}\|_\infty \leq R$, the unbiased sign operation $S_R(\mathbf{v})$ is defined as:

$$[S_R(\mathbf{v})]_k = \begin{cases} 1, & \text{with probability } \frac{R + [\mathbf{v}]_k}{2R}, \\ -1, & \text{with probability } \frac{R - [\mathbf{v}]_k}{2R}. \end{cases} \quad (3)$$

Remark: This operation provides an unbiased estimation of $\frac{\mathbf{v}}{R}$, such that $\mathbb{E}[S_R(\mathbf{v})] = \frac{\mathbf{v}}{R}$.

Note that the unbiased sign operation requires the input vector to be bounded. Therefore, we have to assume the following bounded gradients condition in this section, which is

Algorithm 3 Communication-efficient Distributed Lion Optimizer

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1: Input: Momentum parameters  $\beta_1, \beta_2$ , learning rate  $\eta$ , weight decay parameter  $\lambda$ .
2: for time step  $t = 1$  to  $T$  do
3:   for node  $j = 1$  to  $n$  do
4:     Compute  $\mathbf{v}_t^j = (1 - \beta_1)\mathbf{m}_{t-1}^j + \beta_1 \nabla f_j(\mathbf{x}_t; \xi_t^j)$ 
5:     (v1) Update  $\mathbf{m}_t^j = (1 - \beta_2)\mathbf{m}_{t-1}^j + \beta_2 \nabla f_j(\mathbf{x}_t; \xi_t^j)$ 
6:     (v2) Update  $\mathbf{m}_t^j = (1 - \beta_2)\mathbf{m}_{t-1}^j + \beta_2 \nabla f_j(\mathbf{x}_t; \xi_t^j) + \gamma (\nabla f_j(\mathbf{x}_t; \xi_t^j) - \nabla f_j(\mathbf{x}_{t-1}; \xi_t^j))$ 
7:     Compute  $\mathbf{v}_t = \frac{1}{n} \sum_{j=1}^n Q_1(\mathbf{v}_t^j)$  # Collect and share to all nodes
8:     Update  $\mathbf{x}_{t+1} = \mathbf{x}_t - \eta(Q_2(\mathbf{v}_t) + \lambda \mathbf{x}_t)$ 
9:   end for
10: end for
    
```

also used in previous sign-based distributed algorithms (Jin et al., 2021; Sun et al., 2023).

Assumption 5.2. For each node $j \in [n]$, we assume that the gradient of the loss function is bounded, i.e.,

$$\sup_{\mathbf{x}} \|\nabla f_j(\mathbf{x}; \xi)\|_{\infty} \leq G.$$

Remark: It is important to emphasize that this assumption is needed only to construct the unbiased sign operator $S_R(\mathbf{v})$. This similar sign operator has also been widely used in prior work on distributed bidirectional sign optimization (Jin et al., 2021; Sun et al., 2023), where bounded gradients are also assumed. However, we can also incorporate the more general compression methods (Philippenko & Dieuleveut, 2021; Gruntkowska et al., 2023) to remove this assumption.

Next, we introduce the proposed algorithm. Most steps are similar to those in Algorithm 2, with the key difference being in the parameter update, where we now use:

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \eta \left(Q_2 \left(\frac{1}{n} \sum_{j=1}^n Q_1(\mathbf{v}_t^j) \right) + \lambda \mathbf{x}_t \right),$$

where Q_1 and Q_2 represent the sign or unbiased sign operation, which may vary depending on the specific theorem. The complete algorithm is described in Algorithm 3 (v1). Note that, for the first iteration ($t = 1$), we initialize as $\mathbf{v}_1^j = \mathbf{m}_1^j = \nabla f_j(\mathbf{x}_1; \xi_1^j)$.

Now, we present the convergence guarantees for the proposed algorithm. In the first version, we use the traditional sign operation on the server, i.e., $Q_2(\cdot) = \text{sign}(\cdot)$, and employ the unbiased sign on each node, such that $Q_1(\cdot) = S_G(\cdot)$, where $S_G(\cdot)$ equals $S_R(\cdot)$ with $R = G$. In this case, we establish the following convergence results.

Theorem 5.3. Under Assumptions 4.1, 4.3, 4.4 and 5.2, by setting $\beta_2^2 \leq \beta_1 \leq \sqrt{\beta_2}$, $\beta_2 = \frac{1}{2}$, $Q_1(\cdot) = S_G(\cdot)$, and $Q_2(\cdot) = \text{sign}(\cdot)$, we can obtain the following convergence guarantees:

- Setting $\eta = \mathcal{O}(T^{-1/2}d^{-1/2})$, $\lambda = \mathcal{O}(d^{1/2}T^{-1/2})$, and $\|\mathbf{x}_1\|_{\infty} \leq \mathcal{O}(\lambda^{-1})$, Algorithm 3 (v1) ensures:

$$\begin{aligned} & \frac{1}{T} \sum_{t=1}^T \mathbb{E} [\|\nabla f(\mathbf{x}_t)\|_1] \\ & \leq \mathcal{O} \left(\frac{(1 + \Delta_f + L)d^{1/2}}{T^{1/2}} + \frac{(1 + G + \sigma)d}{n^{1/2}} \right). \end{aligned}$$

- Setting $\eta = \mathcal{O}(n^{-1/2})$, $\lambda = \mathcal{O}(n^{1/2}T^{-1})$, $\|\mathbf{x}_1\|^2 \leq T$ and $\|\mathbf{x}_1\|_{\infty} \leq \mathcal{O}(\lambda^{-1})$, Algorithm 3 (v1) ensures:

$$\begin{aligned} & \frac{1}{T} \sum_{t=1}^T \mathbb{E} [\|\nabla f(\mathbf{x}_t)\|_1] \\ & \leq \mathcal{O} \left(\frac{(1 + \Delta_f + L)n^{1/2}}{T} + \frac{(1 + G + \sigma + L)d}{n^{1/2}} \right). \end{aligned}$$

Remark: By adjusting the learning rate, we obtain the two convergence guarantees above. The second one is preferable when $n \leq Td$. This theorem shows that the gradient l_1 -norm can quickly converge to $\mathcal{O}(\frac{d}{n^{1/2}})$.

However, the above rate has an error floor of $\mathcal{O}(\frac{d}{n^{1/2}})$ and can not converge to zero as T increases. In many scenarios, we would like the upper bound on the averaged gradient norm to decrease to zero as $T \rightarrow \infty$. To achieve this, we further apply an unbiased sign operation on the server, setting $Q_2(\cdot) = S_1(\cdot)$, where $S_1(\cdot)$ equals $S_R(\cdot)$ with $R = 1$. Note that the $S_1(\cdot)$ operation is valid because the input is the average of the sign information $\frac{1}{n} \sum_{j=1}^n Q_1(\mathbf{v}_t^j)$, which lies within $[-1, 1]$. With this modification, we obtain the following convergence guarantee.

Theorem 5.4. Under Assumptions 4.1, 4.3, 4.4 and 5.2, by setting that $\eta = \mathcal{O} \left(\min \left\{ \frac{1}{T^{1/2}d^{1/2}}, \frac{n^{2/5}}{T^{3/5}d^{1/5}} \right\} \right)$, $\beta_2^2 \leq \beta_1 \leq \sqrt{\beta_2}$, $\beta_2 = \mathcal{O}(n^{1/3}\eta^{2/3}d^{1/3})$, $\lambda \leq \mathcal{O} \left(\frac{\sqrt{L}}{T\sqrt{\eta G}} \right)$, $Q_1(\cdot) = S_G(\cdot)$, $Q_2(\cdot) = S_1(\cdot)$, $T \geq n$, and $\|\mathbf{x}_1\|_{\infty} \leq \eta T$,

Algorithm 3 (v1) ensures:

$$\begin{aligned} & \frac{1}{T} \sum_{t=1}^T \mathbb{E} [\|\nabla f(\mathbf{x}_t)\|_1] \\ & \leq \mathcal{O} \left(\sqrt{LG + L^2 + \sigma^2 + \Delta_f G} \right) \max \left\{ \frac{d^{1/4}}{T^{1/4}}, \frac{d^{1/10}}{n^{1/5} T^{1/5}} \right\}. \end{aligned}$$

Remark: The above rate can converge to zero as the iteration number $T \rightarrow \infty$, ensuring that the gradient norm becomes smaller as T increases.

Next, we apply the variance reduction technique along with the average smoothness assumption to further improve the convergence rate. The only modification is that we use the STORM estimator $\mathbf{m}_t^i$ in step 6. The updated algorithm is described in Algorithm 3 (v2). We now present the convergence guarantee for this improved method.

Theorem 5.5. *Under Assumptions 4.2, 4.3, 4.4 and 5.2, setting $\beta_2 \leq \beta_1 \leq \sqrt{\beta_2}$, $\beta_2 = \mathcal{O}(\frac{1}{T^{1/2}})$, $\gamma = 1 - \beta_2$, $\eta = \mathcal{O}(\frac{1}{d^{1/2} T^{1/2}})$, $\lambda = \mathcal{O}(d^{1/4} T^{-3/4})$, $Q_1(\cdot) = S_G(\cdot)$, $Q_2(\cdot) = S_1(\cdot)$, and that $\|\mathbf{x}_1\|_\infty \leq \mathcal{O}(d^{-1/2} T^{1/2})$, Algorithm 3 (v2) ensures:*

$$\begin{aligned} & \frac{1}{T} \sum_{t=1}^T \mathbb{E} [\|\nabla f(\mathbf{x}_t)\|_2] \\ & \leq \mathcal{O} \left(\sqrt{LG + L^2 + \sigma^2 + \Delta_f G} \cdot \frac{d^{1/4}}{T^{1/4}} \right). \end{aligned}$$

Comparison: To ensure $\frac{1}{T} \sum_{t=1}^T \mathbb{E} [\|\nabla f(\mathbf{x}_t)\|_2] \leq \epsilon$, our analysis indicates a sample complexity of $\mathcal{O}(d\epsilon^{-4})$. Compared with other sign-based methods, Jin et al. (2021) require a sample complexity of $\mathcal{O}(d^3\epsilon^{-8})$ and Sun et al. (2023) obtain the complexity $\mathcal{O}(d^4\epsilon^{-4})$ with requiring $n = \mathcal{O}(d^2\epsilon^{-2})$. Furthermore, by directly assuming the existence of an unbiased sign operator C , Gruntkowska et al. (2023) obtain the rate $\mathcal{O}(\omega\epsilon^{-4})$ without requiring the bounded gradient. If we also assume this operator C , we can replace the sign operator $S_R(\mathbf{v})$ in our paper with such C . For example, if we use $\mathbf{x}_{t+1} = \mathbf{x}_t - \eta \left(C \left(\frac{1}{n} \sum_{j=1}^n C(\mathbf{v}_t^j) \right) + \lambda \mathbf{x}_t \right)$, in our Algorithm 3, we ensure the same $\mathcal{O}(d\epsilon^{-4})$ convergence rate for Theorem 5.5. In this case, since $S_R(\mathbf{v})$ is no longer needed, the bounded-gradient assumption can be removed directly. Also, one advantage of this combination is that we do not require the $\mathbb{E}[\|C(\mathbf{x}) - \mathbf{x}\|^2] \leq \omega \|\mathbf{x}\|^2$ assumption as required by Gruntkowska et al. (2023), which can include more unbiased compressors.

6. Experiments

In this section, we empirically validate the effectiveness of the proposed methods in both centralized and distributed settings. Specifically, we evaluate the performance of our

algorithms on image classification tasks using the ImageNet dataset (Deng et al., 2009). All experiments are conducted on eight NVIDIA GeForce RTX 3090 GPUs.

6.1. Centralized setting

We train ResNet-50 (He et al., 2016) and ViT-S (Dosovitskiy et al., 2021) models on the ImageNet dataset. For the ViT-S architecture, we also evaluate a variant that incorporates MixUp (Zhang et al., 2018) and RandAug (Cubuk et al., 2020) data augmentations. We compare our proposed methods with the original Lion optimizer (Chen et al., 2023) and AdamW (Loshchilov & Hutter, 2019), which is widely used as the standard optimizer for many tasks. For hyperparameter tuning, we search the momentum parameters within the range $\{1e-1, 1e-2, 1e-3\}$ (Note that our notation β_1 and β_2 correspond to $1 - \beta_1$ and $1 - \beta_2$ for Adam and the original Lion. This is equivalent to searching β_1 and β_2 in $\{0.9, 0.99, 0.999\}$ with traditional notations). The learning rate is searched from the set $\{1e-2, 3e-3, 1e-3, 3e-4, 1e-4\}$ and the weight decay parameter is tuned within $\{1, 1e-1, 1e-2\}$.

The results are reported in Table 2 and Figure 1. Our Lion-VR method outperforms baselines across all configurations. Specifically, Lion-VR improves upon AdamW (+0.44%) and Lion (+0.46%) on ResNet-50. For ViT-S without augmentation, Lion-VR achieves a 0.33% improvement over Lion and a 0.46% improvement over AdamW. When RandAug and MixUp augmentations are applied, Lion-VR maintains its advantage, providing a 0.12% improvement over Lion and a 0.78% improvement over AdamW. These results demonstrate that incorporating variance reduction into the Lion optimizer leads to empirical gains.

Table 2. Accuracy (%) on ImageNet in the centralized setting.

Model	Optimizer	RandAug +MixUp	Accuracy(%)
ResNet-50	AdamW		76.03±0.31
	Lion	–	76.01±0.22
	Lion-VR		76.47±0.16
ViT-S	AdamW		75.88±0.11
	Lion	✗	76.01±0.09
	Lion-VR		76.34±0.10
	AdamW		78.35±0.06
	Lion	✓	79.01±0.08
	Lion-VR		79.13±0.09

6.2. Distributed setting

In the distributed setting, we train ResNet and ViT models across 4 nodes. We compare these algorithms: (1) the

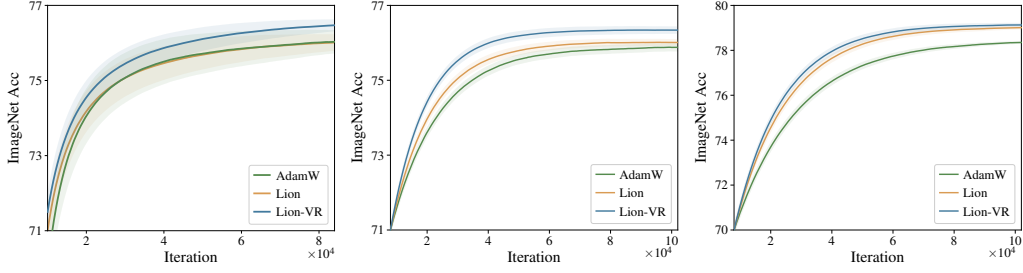


Figure 1. Accuracy curves on ImageNet in the centralized setting for ResNet-50 (left), ViT-S without data augmentation (middle), and ViT-S with RandAug+MixUp (right).

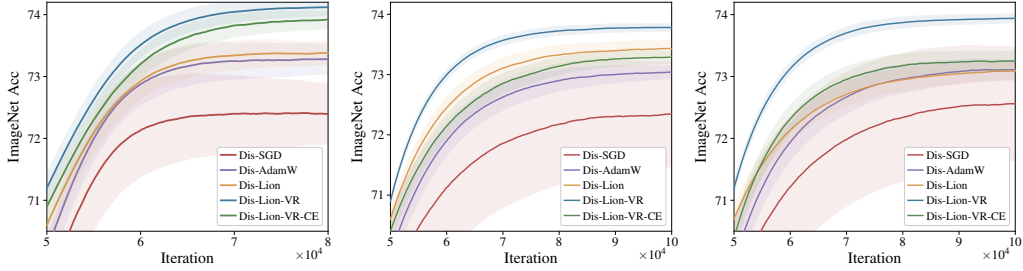


Figure 2. Accuracy curves on ImageNet in the distributed setting for ResNet-50 (left), ViT-S without data augmentation (middle), and ViT-S with RandAug+MixUp (right).

baseline distributed SGD (Dis-SGD) and AdamW (Dis-AdamW), (2) distributed Lion (Dis-Lion, Algorithm 2-v1), (3) our variance-reduced variant (Dis-Lion-VR, Algorithm 2-v2), and (4) its communication-efficient counterpart with bidirectional sign compression (Dis-Lion-VR-CE, Algorithm 3-v2). Hyperparameters are tuned using the same search grids as in the centralized setting.

Table 3. Accuracy (%) on ImageNet in the distributed setting.

Model	Optimizer	RandAug+MixUp	Accuracy (%)
ResNet-50	Dis-SGD	–	72.40±0.49
	Dis-AdamW		73.28±0.25
	Dis-Lion		73.38±0.19
	Dis-Lion-VR		74.12±0.14
	Dis-Lion-VR-CE		73.92±0.15
ViT-S	Dis-SGD	✗	72.35±0.88
	Dis-AdamW		73.04±0.11
	Dis-Lion		73.43±0.14
	Dis-Lion-VR		73.78±0.07
	Dis-Lion-VR-CE		73.29±0.11
	Dis-SGD	✓	72.56±0.93
	Dis-AdamW		73.11±0.16
	Dis-Lion		73.09±0.16
	Dis-Lion-VR		73.94±0.08
	Dis-Lion-VR-CE		73.25±0.16

As shown in Table 3 and Figure 2, Dis-Lion-VR consistently achieves the highest accuracy across all configurations. On

ResNet-50, it improves upon Dis-Lion by 0.74%, and its communication-efficient variant retains most of this gain with a 0.54% improvement. For ViT-S without augmentation, Dis-Lion-VR yields a 0.35% improvement over the baseline. With RandAug and MixUp, this advantage increases to 0.85%. Although Dis-Lion-VR-CE shows lower accuracy compared to Dis-Lion-VR, it maintains competitive performance while substantially reducing communication overhead through bidirectional 1-bit compression.

7. Conclusion

This paper investigates the convergence guarantees of the Lion optimizer in both centralized and distributed settings. We first demonstrate that the Lion optimizer and its variance-reduced version achieve convergence rates of $\mathcal{O}(d^{1/2}T^{-1/4})$ and $\mathcal{O}(d^{1/2}T^{-1/3})$, respectively. We then extend these results to their distributed versions, showing that similar convergence guarantees hold in distributed settings with the linear speedup effect. To further improve communication efficiency, we develop the distributed Lion optimizer with sign compression in both directions, yielding convergence rates of $\mathcal{O}\left(\frac{d^{1/2}}{T^{1/2}} + \frac{d}{n^{1/2}}\right)$ and $\mathcal{O}\left(\frac{n^{1/2}}{T} + \frac{d}{n^{1/2}}\right)$. Finally, by employing unbiased sign operations in both the parameter server and local nodes, we achieve convergence rates of $\mathcal{O}\left(\max\left\{\frac{d^{1/4}}{T^{1/4}}, \frac{d^{1/10}}{n^{1/5}T^{1/5}}\right\}\right)$ and $\mathcal{O}\left(\frac{d^{1/4}}{T^{1/4}}\right)$, respectively. Experiments on image classification tasks further validate the effectiveness of the proposed methods.

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Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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A. Proof of Lemma 3.5

For the update step $\mathbf{x}_{t+1} = \mathbf{x}_t - \eta (\text{sign}(\mathbf{v}_t) + \lambda \mathbf{x}_t)$, we can ensure that

$$\begin{aligned}\|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 &= \eta^2 \|\text{sign}(\mathbf{v}_t) + \lambda \mathbf{x}_t\|^2 \\ &\leq 2\eta^2 \|\text{sign}(\mathbf{v}_t)\|^2 + 2\eta^2 \|\lambda \mathbf{x}_t\|^2 \\ &\leq 2\eta^2 d + 2\eta^2 \lambda^2 \|\mathbf{x}_t\|^2.\end{aligned}$$

Additionally, by Cauchy-Schwarz, we can ensure that

$$\begin{aligned}\|\mathbf{x}_t\|^2 &= \|(1 - \eta\lambda)\mathbf{x}_{t-1} - \eta \text{sign}(\mathbf{v}_{t-1})\|^2 \\ &\leq (1 + \eta\lambda) \|(1 - \eta\lambda)\mathbf{x}_{t-1}\|^2 + (1 + \frac{1}{\eta\lambda}) \|\eta \text{sign}(\mathbf{v}_{t-1})\|^2 \\ &\leq (1 + \eta\lambda)(1 - \eta\lambda)^2 \|\mathbf{x}_{t-1}\|^2 + (1 + \frac{1}{\eta\lambda}) \eta^2 \|\text{sign}(\mathbf{v}_{t-1})\|^2 \\ &\leq (1 - \eta\lambda) \|\mathbf{x}_{t-1}\|^2 + \frac{2\eta^2 d}{\eta\lambda}.\end{aligned}$$

By recursion, this yields

$$\|\mathbf{x}_t\|^2 \leq (1 - \eta\lambda)^{t-1} \|\mathbf{x}_1\|^2 + \frac{2\eta^2 d}{\eta\lambda} \cdot \frac{1 - (1 - \eta\lambda)^t}{1 - (1 - \eta\lambda)} \leq (1 - \eta\lambda)^{t-1} \|\mathbf{x}_1\|^2 + \frac{2\eta^2 d}{(\eta\lambda)^2},$$

and therefore

$$\frac{1}{T} \sum_{t=1}^T \|\mathbf{x}_t\|^2 \leq \frac{1}{T} \sum_{t=1}^T (1 - \eta\lambda)^{t-1} \|\mathbf{x}_1\|^2 + \frac{2\eta^2 d}{(\eta\lambda)^2} \leq \frac{1}{\eta\lambda T} \|\mathbf{x}_1\|^2 + \frac{2\eta^2 d}{(\eta\lambda)^2}.$$

Substituting this bound back, we obtain

$$\begin{aligned}\frac{1}{T} \sum_{t=1}^T \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 &\leq 2\eta^2 d + 2\eta^2 \lambda^2 \frac{1}{T} \sum_{t=1}^T \|\mathbf{x}_t\|^2 \\ &\leq 2\eta^2 d + 2\eta^2 \lambda^2 \left(\frac{1}{\eta\lambda T} \|\mathbf{x}_1\|^2 + \frac{2\eta^2 d}{(\eta\lambda)^2} \right) \leq 6\eta^2 d + \frac{2\eta\lambda \|\mathbf{x}_1\|^2}{T}.\end{aligned}$$

This completes the proof of Lemma 3.5.

B. Proof of Theorem 3.6

Due to the smoothness assumption (Assumption 3.1), we have

$$\begin{aligned}
 & f(\mathbf{x}_{t+1}) \\
 & \leq f(\mathbf{x}_t) + \langle \nabla f(\mathbf{x}_t), \mathbf{x}_{t+1} - \mathbf{x}_t \rangle + L \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2/2 \\
 & = f(\mathbf{x}_t) - \eta \langle \nabla f(\mathbf{x}_t), \text{sign}(\mathbf{v}_t) \rangle - \eta \lambda \langle \nabla f(\mathbf{x}_t), \mathbf{x}_t \rangle + L \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2/2 \\
 & \leq f(\mathbf{x}_t) + \eta \langle \nabla f(\mathbf{x}_t), \text{sign}(\nabla f(\mathbf{x}_t)) - \text{sign}(\mathbf{v}_t) \rangle - \eta \langle \nabla f(\mathbf{x}_t), \text{sign}(\nabla f(\mathbf{x}_t)) \rangle - \eta \lambda \langle \nabla f(\mathbf{x}_t), \mathbf{x}_t \rangle + L \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2/2 \\
 & = f(\mathbf{x}_t) + \eta \langle \nabla f(\mathbf{x}_t), \text{sign}(\nabla f(\mathbf{x}_t)) - \text{sign}(\mathbf{v}_t) \rangle - \eta \|\nabla f(\mathbf{x}_t)\|_1 + \eta \lambda \|\nabla f(\mathbf{x}_t)\|_1 \|\mathbf{x}_t\|_\infty + L \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2/2 \\
 & \leq f(\mathbf{x}_t) + \eta \sum_{i=1}^d \langle [\nabla f(\mathbf{x}_t)]_i, \text{sign}([\nabla f(\mathbf{x}_t)]_i) - \text{sign}([\mathbf{v}_t]_i) \rangle - \frac{\eta}{2} \|\nabla f(\mathbf{x}_t)\|_1 + L \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2/2 \\
 & \leq f(\mathbf{x}_t) + \eta \sum_{i=1}^d 2|[\nabla f(\mathbf{x}_t)]_i - [\mathbf{v}_t]_i| \cdot \mathbb{I}(\text{sign}([\nabla f(\mathbf{x}_t)]_i) \neq \text{sign}([\mathbf{v}_t]_i)) - \frac{\eta}{2} \|\nabla f(\mathbf{x}_t)\|_1 + L \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2/2 \\
 & \leq f(\mathbf{x}_t) + \eta \sum_{i=1}^d 2|[\nabla f(\mathbf{x}_t)]_i - [\mathbf{v}_t]_i| - \frac{\eta}{2} \|\nabla f(\mathbf{x}_t)\|_1 + L \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2/2 \\
 & = f(\mathbf{x}_t) + 2\eta \|\nabla f(\mathbf{x}_t) - \mathbf{v}_t\|_1 - \frac{\eta}{2} \|\nabla f(\mathbf{x}_t)\|_1 + L \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2/2 \\
 & \leq f(\mathbf{x}_t) + 2\eta\sqrt{d} \|\nabla f(\mathbf{x}_t) - \mathbf{v}_t\| - \frac{\eta}{2} \|\nabla f(\mathbf{x}_t)\|_1 + L \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2/2,
 \end{aligned} \tag{4}$$

where the third inequality is valid with $\|\mathbf{x}_1\|_\infty \leq \eta T$ and $4\lambda\eta T \leq 1$, such that

$$\begin{aligned}
 \|\mathbf{x}_t\|_\infty & = \|(1 - \eta\lambda)\mathbf{x}_{t-1} - \eta \text{sign}(\mathbf{v}_{t-1})\|_\infty \leq (1 - \eta\lambda) \|\mathbf{x}_{t-1}\|_\infty + \eta \|\text{sign}(\mathbf{v}_{t-1})\|_\infty \\
 & \leq (1 - \eta\lambda)^{t-1} \|\mathbf{x}_1\|_\infty + \eta T \leq 2\eta T.
 \end{aligned}$$

Summing up and rearranging equation (4), we derive:

$$\begin{aligned}
 \mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t)\|_1 \right] & \leq \frac{2(f(\mathbf{x}_1) - f(\mathbf{x}_{T+1}))}{\eta T} + 4\sqrt{d} \cdot \mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t) - \mathbf{v}_t\| \right] + \frac{L}{\eta T} \sum_{t=1}^T \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 \\
 & \leq \frac{2(\Delta_f + L)}{\eta T} + 4\sqrt{d} \cdot \sqrt{\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t) - \mathbf{v}_t\|^2 \right]} + 6\eta Ld,
 \end{aligned} \tag{5}$$

where $f(\mathbf{x}_1) - f_* \leq \Delta_f$, and the second inequality is due to Jensen's Inequality and setting $\lambda\eta \|\mathbf{x}_1\|^2 \leq 1$.

Next, we bound the term $\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t) - \mathbf{v}_t\|^2 \right]$ as follows.

$$\begin{aligned}
 & \mathbb{E} \left[\|\nabla f(\mathbf{x}_{t+1}) - \mathbf{v}_{t+1}\|^2 \right] \\
 & = \mathbb{E} \left[\|(1 - \beta_1)\mathbf{m}_t + \beta_1 \nabla f(\mathbf{x}_{t+1}; \xi_{t+1}) - \nabla f(\mathbf{x}_{t+1})\|^2 \right] \\
 & = \mathbb{E} \left[\|(1 - \beta_1)(\mathbf{m}_t - \nabla f(\mathbf{x}_t)) + \beta_1 (\nabla f(\mathbf{x}_{t+1}; \xi_{t+1}) - \nabla f(\mathbf{x}_{t+1})) + (1 - \beta_1) (\nabla f(\mathbf{x}_t) - \nabla f(\mathbf{x}_{t+1}))\|^2 \right] \\
 & = (1 - \beta_1)^2 \mathbb{E} \left[\|\mathbf{m}_t - \nabla f(\mathbf{x}_t) + \nabla f(\mathbf{x}_t) - \nabla f(\mathbf{x}_{t+1})\|^2 \right] + \beta_1^2 \mathbb{E} \left[\|\nabla f(\mathbf{x}_{t+1}; \xi_{t+1}) - \nabla f(\mathbf{x}_{t+1})\|^2 \right] \\
 & \leq (1 - \beta_1)^2 (1 + \beta_1) \mathbb{E} \left[\|\mathbf{m}_t - \nabla f(\mathbf{x}_t)\|^2 \right] + (1 - \beta_1)^2 (1 + \frac{1}{\beta_1}) \mathbb{E} \left[\|\nabla f(\mathbf{x}_t) - \nabla f(\mathbf{x}_{t+1})\|^2 \right] + \beta_1^2 \sigma^2 \\
 & \leq (1 - \beta_1) \mathbb{E} \left[\|\mathbf{m}_t - \nabla f(\mathbf{x}_t)\|^2 \right] + \frac{2L^2}{\beta_1} \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 + \beta_1^2 \sigma^2 \\
 & \leq (1 - \beta_1) \mathbb{E} \left[\|\mathbf{m}_t - \nabla f(\mathbf{x}_t)\|^2 \right] + \frac{8\eta^2 L^2 d}{\beta_1} + \beta_1^2 \sigma^2,
 \end{aligned} \tag{6}$$

where the third equality is due to the fact $\mathbb{E} [\nabla f(\mathbf{x}_{t+1}; \xi_{t+1}) - \nabla f(\mathbf{x}_{t+1})] = 0$.

Very similarly, we can obtain that

$$\begin{aligned} \mathbb{E} [\|\nabla f(\mathbf{x}_{t+1}) - \mathbf{m}_{t+1}\|^2] &= \mathbb{E} [\|(1 - \beta_2)\mathbf{m}_t + \beta_2 \nabla f(\mathbf{x}_{t+1}; \xi_{t+1}) - \nabla f(\mathbf{x}_{t+1})\|^2] \\ &\leq (1 - \beta_2) \mathbb{E} [\|\mathbf{m}_t - \nabla f(\mathbf{x}_t)\|^2] + \frac{8\eta^2 L^2 d}{\beta_2} + \beta_2^2 \sigma^2. \end{aligned}$$

Summing up, we observe

$$\begin{aligned} \mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\mathbf{m}_t - \nabla f(\mathbf{x}_t)\|^2 \right] &\leq \frac{\mathbb{E} [\|\mathbf{m}_1 - \nabla f(\mathbf{x}_1)\|^2]}{\beta_2 T} + \frac{8\eta^2 L^2 d}{\beta_2^2} + \beta_2 \sigma^2 \\ &\leq \frac{\sigma^2}{\beta_2 T} + \frac{8\eta^2 L^2 d}{\beta_2^2} + \beta_2 \sigma^2. \end{aligned} \tag{7}$$

Thus, by setting $\beta_2^2 \leq \beta_1 \leq \sqrt{\beta_2}$, we can ensure

$$\begin{aligned} \mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\mathbf{v}_t - \nabla f(\mathbf{x}_t)\|^2 \right] &\leq \frac{\sigma^2}{T} + (1 - \beta_1) \mathbb{E} \left[\frac{1}{T} \sum_{t=1}^{T-1} \|\mathbf{m}_t - \nabla f(\mathbf{x}_t)\|^2 \right] + \frac{8\eta^2 L^2 d}{\beta_1} + \beta_1^2 \sigma^2 \\ &\leq \frac{\sigma^2}{T} + \frac{\sigma^2}{\beta_2 T} + \frac{8\eta^2 L^2 d}{\beta_2^2} + \beta_2 \sigma^2 + \frac{8\eta^2 L^2 d}{\beta_1} + \beta_1^2 \sigma^2 \\ &\leq \frac{2\sigma^2}{\beta_2 T} + \frac{16\eta^2 L^2 d}{\beta_2^2} + 2\beta_2 \sigma^2. \end{aligned}$$

By setting $\beta_2 = \mathcal{O}(T^{-1/2})$, $\eta = \mathcal{O}(d^{-1/2}T^{-3/4})$, we obtain:

$$\begin{aligned} \mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t)\|_1 \right] &\leq \frac{2(\Delta_f + L)}{\eta T} + 4\sqrt{d} \cdot \sqrt{\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t) - \mathbf{v}_t\|^2 \right]} + 6\eta L d \\ &\leq \frac{2(\Delta_f + L)}{\eta T} + 4\sqrt{d} \cdot \sqrt{\frac{2\sigma^2}{\beta_2 T} + \frac{16\eta^2 L^2 d}{\beta_2^2} + 2\beta_2 \sigma^2} + 6\eta L d \\ &= \mathcal{O} \left(\frac{(1 + \Delta_f + \sigma + L) d^{1/2}}{T^{1/4}} \right) \\ &= \mathcal{O} \left(\frac{d^{1/2}}{T^{1/4}} \right). \end{aligned}$$

This completes the proof of Theorem 3.6.

C. Proof of Theorem 3.7

Since the variance reduction version only differs from the original version in step 5, the analysis of Theorem 3.6 holds up until equation (6).

The main difference is that we can bound the estimator error $\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t) - \mathbf{m}_t\|^2 \right]$ as follows:

$$\begin{aligned}
 & \mathbb{E} \left[\|\nabla f(\mathbf{x}_{t+1}) - \mathbf{m}_{t+1}\|^2 \right] \\
 &= \mathbb{E} \left[\|(1 - \beta_2)\mathbf{m}_t + \beta_2 \nabla f(\mathbf{x}_{t+1}; \xi_{t+1}) + (1 - \beta_2)(\nabla f(\mathbf{x}_{t+1}; \xi_{t+1}) - \nabla f(\mathbf{x}_t; \xi_{t+1})) - \nabla f(\mathbf{x}_{t+1})\|^2 \right] \\
 &= \mathbb{E} \left[\|(1 - \beta_2)(\mathbf{m}_t - \nabla f(\mathbf{x}_t)) + \beta_2 (\nabla f(\mathbf{x}_t; \xi_{t+1}) - \nabla f(\mathbf{x}_t)) \right. \\
 &\quad \left. + (\nabla f(\mathbf{x}_t) - \nabla f(\mathbf{x}_{t+1}) + \nabla f(\mathbf{x}_{t+1}; \xi_{t+1}) - \nabla f(\mathbf{x}_t; \xi_{t+1}))\|^2 \right] \\
 &\leq (1 - \beta_2)^2 \mathbb{E} \left[\|\mathbf{m}_t - \nabla f(\mathbf{x}_t)\|^2 \right] + 2\beta_2^2 \mathbb{E} \left[\|\nabla f(\mathbf{x}_t; \xi_{t+1}) - \nabla f(\mathbf{x}_t)\|^2 \right] \\
 &\quad + 2\mathbb{E} \left[\|(\nabla f(\mathbf{x}_{t+1}; \xi_{t+1}) - \nabla f(\mathbf{x}_t; \xi_{t+1}))\|^2 \right] \\
 &\leq (1 - \beta_2) \mathbb{E} \left[\|\mathbf{m}_t - \nabla f(\mathbf{x}_t)\|^2 \right] + 2\beta_2^2 \sigma^2 + 2L^2 \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 \\
 &\leq (1 - \beta_2) \mathbb{E} \left[\|\mathbf{m}_t - \nabla f(\mathbf{x}_t)\|^2 \right] + 2\beta_2^2 \sigma^2 + 8L^2 \eta^2 d,
 \end{aligned}$$

where the first inequality is due to the fact that $(a + b)^2 \leq 2a^2 + 2b^2$, and the fact that

$$\mathbb{E} \left[(\beta (\nabla f(\mathbf{x}_t; \xi_{t+1}) - \nabla f(\mathbf{x}_t)) + \nabla f(\mathbf{x}_t) - \nabla f(\mathbf{x}_{t+1}) + \nabla f(\mathbf{x}_{t+1}; \xi_{t+1}) - \nabla f(\mathbf{x}_t; \xi_{t+1})) \right] = 0.$$

Summing up, and noting that we use a batch size of B_0 in the first iteration, we can ensure that

$$\begin{aligned}
 \mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\mathbf{m}_t - \nabla f(\mathbf{x}_t)\|^2 \right] &\leq \frac{\mathbb{E} \left[\|\mathbf{m}_1 - \nabla f(\mathbf{x}_1)\|^2 \right]}{\beta_2 T} + 2\sigma^2 \beta_2 + \frac{8L^2 \eta^2 d}{\beta_2} \\
 &\leq \frac{\sigma^2}{B_0 \beta_2 T} + 2\sigma^2 \beta_2 + \frac{8L^2 \eta^2 d}{\beta_2}.
 \end{aligned} \tag{8}$$

Thus, by setting $\beta_2 \leq \beta_1 \leq \sqrt{\beta_2}$ and $B_0 \beta_2 \leq 1$, we can ensure

$$\begin{aligned}
 \mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\mathbf{v}_t - \nabla f(\mathbf{x}_t)\|^2 \right] &\leq \frac{\sigma^2}{T} + (1 - \beta_1) \mathbb{E} \left[\frac{1}{T} \sum_{t=1}^{T-1} \|\mathbf{m}_t - \nabla f(\mathbf{x}_t)\|^2 \right] + \frac{8\eta^2 L^2 d}{\beta_1} + \beta_1^2 \sigma^2 \\
 &\leq \frac{\sigma^2}{T} + \frac{\sigma^2}{B_0 \beta_2 T} + 2\beta_2 \sigma^2 + \frac{8\eta^2 L^2 d}{\beta_2} + \frac{8\eta^2 L^2 d}{\beta_1} + \beta_1^2 \sigma^2 \\
 &\leq \frac{2\sigma^2}{B_0 \beta_2 T} + \frac{16\eta^2 L^2 d}{\beta_2} + 3\beta_2 \sigma^2.
 \end{aligned}$$

By setting $\beta_2 = \mathcal{O}(T^{-2/3})$, $\eta = \mathcal{O}(d^{-1/2} T^{-2/3})$, and $B_0 = \mathcal{O}(T^{1/3})$, we observe:

$$\begin{aligned}
 \mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t)\|_1 \right] &\leq \frac{2(\Delta_f + L)}{\eta T} + 4\sqrt{d} \cdot \sqrt{\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t) - \mathbf{v}_t\|^2 \right]} + 6\eta L d \\
 &\leq \frac{2(\Delta_f + L)}{\eta T} + 4\sqrt{d} \cdot \sqrt{\frac{2\sigma^2}{B_0 \beta_2 T} + \frac{16\eta^2 L^2 d}{\beta_2} + 3\beta_2 \sigma^2 + 6\eta L d} \\
 &= \mathcal{O} \left(\frac{(1 + \Delta_f + \sigma + L) d^{1/2}}{T^{1/3}} \right) \\
 &= \mathcal{O} \left(\frac{d^{1/2}}{T^{1/3}} \right).
 \end{aligned}$$

This completes the proof of Theorem 3.7.

D. Proof of Theorem 4.5

The analysis of Theorem 3.6 is valid for Theorem 4.5 until equation (5). That is, we can still ensure

$$\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t)\|_1 \right] \leq \frac{2(\Delta_f + L)}{\eta T} + 4\sqrt{d} \cdot \sqrt{\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t) - \mathbf{v}_t\|^2 \right]} + 6\eta Ld.$$

Next, we bound $\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t) - \mathbf{v}_t\|^2 \right] = \mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \left\| \frac{1}{n} \sum_{j=1}^n \mathbf{v}_t^j - \nabla f(\mathbf{x}_t) \right\|^2 \right]$ as follows. For each worker j , we have the following according to the definition of $\mathbf{v}_t^j$:

$$\begin{aligned} \mathbf{v}_{t+1}^j - \nabla f_j(\mathbf{x}_{t+1}) &= (1 - \beta_1) (\mathbf{m}_t^j - \nabla f_j(\mathbf{x}_t)) + \beta_1 (\nabla f_j(\mathbf{x}_{t+1}; \xi_{t+1}^j) - \nabla f_j(\mathbf{x}_{t+1})) \\ &\quad + (1 - \beta_1) (\nabla f_j(\mathbf{x}_t) - \nabla f_j(\mathbf{x}_{t+1})). \end{aligned}$$

Averaging over $\{n\}$ and noting that $\nabla f(\mathbf{x}) = \frac{1}{n} \sum_{j=1}^n \nabla f_j(\mathbf{x})$, we obtain

$$\begin{aligned} \frac{1}{n} \sum_{j=1}^n \mathbf{v}_{t+1}^j - \nabla f(\mathbf{x}_{t+1}) &= \frac{1}{n} \sum_{j=1}^n (\mathbf{v}_{t+1}^j - \nabla f_j(\mathbf{x}_{t+1})) \\ &= (1 - \beta_1) \frac{1}{n} \sum_{j=1}^n (\mathbf{m}_t^j - \nabla f_j(\mathbf{x}_t)) + \beta_1 \frac{1}{n} \sum_{j=1}^n (\nabla f_j(\mathbf{x}_{t+1}; \xi_{t+1}^j) - \nabla f_j(\mathbf{x}_{t+1})) \\ &\quad + (1 - \beta_1) \frac{1}{n} \sum_{j=1}^n (\nabla f_j(\mathbf{x}_t) - \nabla f_j(\mathbf{x}_{t+1})). \end{aligned}$$

Then, we have

$$\begin{aligned} &\mathbb{E} \left[\left\| \frac{1}{n} \sum_{j=1}^n \mathbf{v}_{t+1}^j - \nabla f(\mathbf{x}_{t+1}) \right\|^2 \right] \\ &\leq (1 - \beta_1) \mathbb{E} \left[\left\| \frac{1}{n} \sum_{j=1}^n (\mathbf{m}_t^j - \nabla f_j(\mathbf{x}_t)) \right\|^2 \right] + \beta_1^2 \frac{1}{n^2} \sum_{j=1}^n \mathbb{E} \left[\left\| \nabla f_j(\mathbf{x}_{t+1}; \xi_{t+1}^j) - \nabla f_j(\mathbf{x}_{t+1}) \right\|^2 \right] \\ &\quad + \frac{2}{\beta_1 n} \sum_{j=1}^n \mathbb{E} \left[\left\| \nabla f_j(\mathbf{x}_{t+1}) - \nabla f_j(\mathbf{x}_t) \right\|^2 \right] \\ &\leq (1 - \beta_1) \mathbb{E} \left[\left\| \frac{1}{n} \sum_{j=1}^n (\mathbf{m}_t^j - \nabla f_j(\mathbf{x}_t)) \right\|^2 \right] + \frac{\beta_1^2 \sigma^2}{n} + \frac{2L^2}{\beta_1} \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 \\ &\leq (1 - \beta_1) \mathbb{E} \left[\left\| \frac{1}{n} \sum_{j=1}^n \mathbf{m}_t^j - \nabla f(\mathbf{x}_t) \right\|^2 \right] + \frac{\beta_1^2 \sigma^2}{n} + \frac{2L^2}{\beta_1} \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2. \end{aligned} \tag{9}$$

Very similarly, we obtain

$$\mathbb{E} \left[\left\| \frac{1}{n} \sum_{j=1}^n \mathbf{m}_{t+1}^j - \nabla f(\mathbf{x}_{t+1}) \right\|^2 \right] \leq (1 - \beta_2) \mathbb{E} \left[\left\| \frac{1}{n} \sum_{j=1}^n \mathbf{m}_t^j - \nabla f(\mathbf{x}_t) \right\|^2 \right] + \frac{\beta_2^2 \sigma^2}{n} + \frac{2L^2}{\beta_2} \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2.$$

By summing up and rearranging, we observe

$$\begin{aligned} \mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \left\| \frac{1}{n} \sum_{j=1}^n \mathbf{m}_t^j - \nabla f(\mathbf{x}_t) \right\|^2 \right] &\leq \frac{\mathbb{E} \left[\left\| \frac{1}{n} \sum_{j=1}^n \mathbf{m}_1^j - \nabla f(\mathbf{x}_1) \right\|^2 \right]}{\beta_2 T} + \frac{\beta_2 \sigma^2}{n} + \frac{1}{T} \sum_{t=1}^T \frac{2L^2}{\beta_2^2} \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 \\ &\leq \frac{\sigma^2}{n\beta_2 T} + \frac{\sigma^2 \beta_2}{n} + \frac{12L^2 \eta^2 d}{\beta_2^2} + \frac{4\eta \lambda L^2 \|\mathbf{x}_1\|^2}{\beta_2^2 T}. \end{aligned} \quad (10)$$

Therefore, by setting $\beta_2^2 \leq \beta_1 \leq \sqrt{\beta_2}$ and $\|\mathbf{x}_1\|^2 \leq \eta d T / \lambda$, we know that

$$\begin{aligned} \mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \left\| \frac{1}{n} \sum_{j=1}^n \mathbf{v}_t^j - \nabla f(\mathbf{x}_t) \right\|^2 \right] &\leq \frac{\sigma^2}{nT} + (1 - \beta_1) \mathbb{E} \left[\frac{1}{T} \sum_{t=1}^{T-1} \left\| \frac{1}{n} \sum_{j=1}^n \mathbf{m}_t^j - \nabla f(\mathbf{x}_t) \right\|^2 \right] + \frac{\beta_1^2 \sigma^2}{n} + \frac{2L^2}{\beta_1 T} \sum_{t=1}^T \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 \\ &\leq \frac{\sigma^2}{nT} + \frac{\sigma^2}{n\beta_2 T} + \frac{\sigma^2 \beta_2}{n} + \frac{12L^2 \eta^2 d}{\beta_2^2} + \frac{4\eta \lambda L^2 \|\mathbf{x}_1\|^2}{\beta_2^2 T} + \frac{\beta_1^2 \sigma^2}{n} + \frac{12L^2 \eta^2 d}{\beta_1} + \frac{4L^2 \eta \lambda \|\mathbf{x}_1\|^2}{\beta_1 T} \\ &\leq \frac{2\sigma^2}{\beta_2 n T} + \frac{32\eta^2 L^2 d}{\beta_2^2} + \frac{2\beta_2 \sigma^2}{n}. \end{aligned}$$

By setting $\beta_2 = \mathcal{O}(n^{1/2} T^{-1/2})$, $\eta = \mathcal{O}(n^{1/4} d^{-1/2} T^{-3/4})$, and $T \geq n$, we observe:

$$\begin{aligned} \mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t)\|_1 \right] &\leq \frac{2(\Delta_f + L)}{\eta T} + 4\sqrt{d} \cdot \sqrt{\frac{2\sigma^2}{\beta_2 n T} + \frac{32\eta^2 L^2 d}{\beta_2^2} + \frac{2\beta_2 \sigma^2}{n}} + 6\eta L d \\ &\leq \mathcal{O} \left(\frac{(1 + \Delta_f + \sigma + L) d^{1/2}}{(nT)^{1/4}} \right) \\ &\leq \mathcal{O} \left(\frac{d^{1/2}}{(nT)^{1/4}} \right). \end{aligned}$$

This completes the proof of Theorem 4.5.

E. Proof of Theorem 4.6

The analysis of Theorem 3.6 is also valid for Theorem 4.6 until equation (5). The first difference is that we can bound the term $\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \left\| \frac{1}{n} \sum_{j=1}^n \mathbf{m}_t^j - \nabla f(\mathbf{x}_t) \right\|^2 \right]$ as follows.

For each worker j , we have the following according to the definition of $\mathbf{m}_t^j$:

$$\begin{aligned} \mathbf{m}_{t+1}^j - \nabla f_j(\mathbf{x}_{t+1}) &= (1 - \beta_2) \left(\mathbf{m}_t^j - \nabla f_j(\mathbf{x}_t) \right) + \beta_2 \left(\nabla f_j(\mathbf{x}_{t+1}; \xi_{t+1}^j) - \nabla f_j(\mathbf{x}_{t+1}) \right) \\ &\quad + (1 - \beta_2) \left(\nabla f_j(\mathbf{x}_{t+1}; \xi_{t+1}^j) - \nabla f_j(\mathbf{x}_t; \xi_{t+1}^j) + \nabla f_j(\mathbf{x}_t) - \nabla f_j(\mathbf{x}_{t+1}) \right). \end{aligned}$$

Summing over $\{n\}$ and noting that $\nabla f(\mathbf{x}) = \frac{1}{n} \sum_{j=1}^n \nabla f_j(\mathbf{x})$, we obtain:

$$\begin{aligned} \frac{1}{n} \sum_{j=1}^n \mathbf{m}_{t+1}^j - \nabla f(\mathbf{x}_{t+1}) &= \frac{1}{n} \sum_{j=1}^n \left(\mathbf{m}_t^j - \nabla f_j(\mathbf{x}_t) \right) \\ &= (1 - \beta_2) \frac{1}{n} \sum_{j=1}^n \left(\mathbf{m}_t^j - \nabla f_j(\mathbf{x}_t) \right) + \beta_2 \frac{1}{n} \sum_{j=1}^n \left(\nabla f_j(\mathbf{x}_{t+1}; \xi_{t+1}^j) - \nabla f_j(\mathbf{x}_{t+1}) \right) \\ &\quad + (1 - \beta_2) \frac{1}{n} \sum_{j=1}^n \left(\nabla f_j(\mathbf{x}_{t+1}; \xi_{t+1}^j) - \nabla f_j(\mathbf{x}_t; \xi_{t+1}^j) + \nabla f_j(\mathbf{x}_t) - \nabla f_j(\mathbf{x}_{t+1}) \right). \end{aligned}$$

Thus, we have

$$\begin{aligned} &\mathbb{E} \left[\left\| \frac{1}{n} \sum_{j=1}^n \mathbf{m}_{t+1}^j - \nabla f(\mathbf{x}_{t+1}) \right\|^2 \right] \\ &\leq (1 - \beta_2)^2 \mathbb{E} \left[\left\| \frac{1}{n} \sum_{j=1}^n \left(\mathbf{m}_t^j - \nabla f_j(\mathbf{x}_t) \right) \right\|^2 \right] + 2\beta_2^2 \frac{1}{n^2} \sum_{j=1}^n \mathbb{E} \left[\left\| \nabla f_j(\mathbf{x}_{t+1}; \xi_{t+1}^j) - \nabla f_j(\mathbf{x}_{t+1}) \right\|^2 \right] \\ &\quad + 2(1 - \beta_2)^2 \frac{1}{n^2} \sum_{j=1}^n \mathbb{E} \left[\left\| \nabla f_j(\mathbf{x}_{t+1}; \xi_{t+1}^j) - \nabla f_j(\mathbf{x}_t; \xi_{t+1}^j) \right\|^2 \right] \\ &\leq (1 - \beta_2) \mathbb{E} \left[\left\| \frac{1}{n} \sum_{j=1}^n \left(\mathbf{m}_t^j - \nabla f_j(\mathbf{x}_t) \right) \right\|^2 \right] + \frac{2\beta_2^2 \sigma^2}{n} + \frac{2L^2}{n} \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 \\ &\leq (1 - \beta_2) \mathbb{E} \left[\left\| \frac{1}{n} \sum_{j=1}^n \mathbf{m}_t^j - \nabla f(\mathbf{x}_t) \right\|^2 \right] + \frac{2\beta_2^2 \sigma^2}{n} + \frac{2L^2}{n} \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2. \end{aligned}$$

By summing up and rearranging, we have

$$\begin{aligned} \mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \left\| \frac{1}{n} \sum_{j=1}^n \mathbf{m}_t^j - \nabla f(\mathbf{x}_t) \right\|^2 \right] &\leq \frac{\mathbb{E} \left[\left\| \frac{1}{n} \sum_{j=1}^n \mathbf{m}_1^j - \nabla f(\mathbf{x}_1) \right\|^2 \right]}{\beta_2 T} + \frac{2\sigma^2 \beta_2}{n} + \frac{2L^2}{n\beta_2 T} \sum_{t=1}^T \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 \\ &\leq \frac{\sigma^2}{n\beta_2 B_0 T} + \frac{2\sigma^2 \beta_2}{n} + \frac{16L^2 \eta^2 d}{n\beta_2}, \end{aligned} \tag{11}$$

where B_0 is the batch size in the iteration number. Similarly, we obtain:

$$\mathbb{E} \left[\left\| \frac{1}{n} \sum_{j=1}^n \mathbf{v}_{t+1}^j - \nabla f(\mathbf{x}_{t+1}) \right\|^2 \right] \leq (1 - \beta_1) \mathbb{E} \left[\left\| \frac{1}{n} \sum_{j=1}^n \mathbf{m}_t^j - \nabla f(\mathbf{x}_t) \right\|^2 \right] + \frac{2\beta_1^2 \sigma^2}{n} + \frac{2L^2}{n} \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2.$$

As a result, by setting $\beta_1 \leq \sqrt{\beta_2}$, we know that

$$\begin{aligned}
 \mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \left\| \frac{1}{n} \sum_{j=1}^n \mathbf{v}_t^j - \nabla f(\mathbf{x}_t) \right\|^2 \right] &\leq \frac{\sigma^2}{nT} + (1 - \beta_1) \mathbb{E} \left[\frac{1}{T} \sum_{t=1}^{T-1} \left\| \frac{1}{n} \sum_{j=1}^n \mathbf{m}_t^j - \nabla f(\mathbf{x}_t) \right\|^2 \right] + \frac{2\beta_1^2 \sigma^2}{n} + \frac{2L^2}{nT} \sum_{t=1}^T \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 \\
 &\leq \frac{\sigma^2}{nT} + \frac{\sigma^2}{n\beta_2 B_0 T} + \frac{2\sigma^2 \beta_2}{n} + \frac{16L^2 \eta^2 d}{n\beta_2} + \frac{2\beta_1^2 \sigma^2}{n} + \frac{2L^2}{nT} \sum_{t=1}^T \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 \\
 &\leq \frac{2\sigma^2}{\beta_2 n B_0 T} + \frac{32\eta^2 L^2 d}{n\beta_2} + \frac{4\beta_2 \sigma^2}{n}.
 \end{aligned}$$

Set $\beta_2 = \mathcal{O}(n^{1/3} T^{-2/3})$, $\eta = \mathcal{O}(n^{1/3} d^{-1/2} T^{-2/3})$, $B_0 = \mathcal{O}(n^{-2/3} T^{1/3})$, and $T \geq n^2$, we have:

$$\begin{aligned}
 \mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t)\|_1 \right] &\leq \frac{2(\Delta_f + L)}{\eta T} + 4\sqrt{d} \cdot \sqrt{\frac{2\sigma^2}{\beta_2 n B_0 T} + \frac{32\eta^2 L^2 d}{n\beta_2} + \frac{4\beta_2 \sigma^2}{n}} + 6\eta L d \\
 &\leq \mathcal{O} \left(\frac{(1 + \Delta_f + \sigma + L) d^{1/2}}{(nT)^{1/3}} \right) \\
 &\leq \mathcal{O} \left(\frac{d^{1/2}}{(nT)^{1/3}} \right).
 \end{aligned}$$

This completes the proof of Theorem 4.6.

F. Proof of Theorem 5.3

Since the overall objective function $f(\mathbf{x})$ is L -smooth, we have the following:

$$\begin{aligned}
 f(\mathbf{x}_{t+1}) &\leq f(\mathbf{x}_t) + \langle \nabla f(\mathbf{x}_t), \mathbf{x}_{t+1} - \mathbf{x}_t \rangle + \frac{L}{2} \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 \\
 &\leq f(\mathbf{x}_t) - \eta \left\langle \nabla f(\mathbf{x}_t), \text{sign} \left(\frac{1}{n} \sum_{j=1}^n S_G(\mathbf{v}_t^j) \right) \right\rangle - \eta \lambda \langle \nabla f(\mathbf{x}_t), \mathbf{x}_t \rangle + \frac{L}{2} \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 \\
 &= f(\mathbf{x}_t) + \eta \left\langle \nabla f(\mathbf{x}_t), \text{sign}(\nabla f(\mathbf{x}_t)) - \text{sign} \left(\frac{1}{n} \sum_{j=1}^n S_G(\mathbf{v}_t^j) \right) \right\rangle \\
 &\quad - \eta \langle \nabla f(\mathbf{x}_t), \text{sign}(\nabla f(\mathbf{x}_t)) \rangle + \eta \lambda \|\nabla f(\mathbf{x}_t)\|_1 \|\mathbf{x}_t\|_\infty + \frac{L}{2} \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 \\
 &= f(\mathbf{x}_t) + \eta \left\langle \nabla f(\mathbf{x}_t), \text{sign}(\nabla f(\mathbf{x}_t)) - \text{sign} \left(\frac{1}{n} \sum_{j=1}^n S_G(\mathbf{v}_t^j) \right) \right\rangle - \frac{\eta}{2} \|\nabla f(\mathbf{x}_t)\|_1 + \frac{L}{2} \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 \\
 &\leq f(\mathbf{x}_t) + 2\eta G \sqrt{d} \left\| \frac{\nabla f(\mathbf{x}_t)}{G} - \frac{1}{n} \sum_{j=1}^n S_G(\mathbf{v}_t^j) \right\| - \frac{\eta}{2} \|\nabla f(\mathbf{x}_t)\|_1 + \frac{L}{2} \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2,
 \end{aligned} \tag{12}$$

where the last inequality is because of

$$\begin{aligned}
 &\left\langle \nabla f(\mathbf{x}_t), \text{sign}(\nabla f(\mathbf{x}_t)) - \text{sign} \left(\frac{1}{n} \sum_{j=1}^n S_G(\mathbf{v}_t^j) \right) \right\rangle \\
 &= \sum_{i=1}^d \left\langle [\nabla f(\mathbf{x}_t)]_i, \text{sign}([\nabla f(\mathbf{x}_t)]_i) - \text{sign} \left(\left[\frac{1}{n} \sum_{j=1}^n S_G(\mathbf{v}_t^j) \right]_i \right) \right\rangle \\
 &\leq \sum_{i=1}^d 2 \left| [\nabla f(\mathbf{x}_t)]_i \right| \cdot \mathbb{I} \left(\text{sign}([\nabla f(\mathbf{x}_t)]_i) \neq \text{sign} \left(\left[\frac{1}{n} \sum_{j=1}^n S_G(\mathbf{v}_t^j) \right]_i \right) \right) \\
 &\leq \sum_{i=1}^d 2G \left| \frac{[\nabla f(\mathbf{x}_t)]_i}{G} \right| \cdot \mathbb{I} \left(\text{sign}([\nabla f(\mathbf{x}_t)]_i) \neq \text{sign} \left(\left[\frac{1}{n} \sum_{j=1}^n S_G(\mathbf{v}_t^j) \right]_i \right) \right) \\
 &\leq \sum_{i=1}^d 2G \left| \frac{[\nabla f(\mathbf{x}_t)]_i}{G} - \left[\frac{1}{n} \sum_{j=1}^n S_G(\mathbf{v}_t^j) \right]_i \right| \cdot \mathbb{I} \left(\text{sign}([\nabla f(\mathbf{x}_t)]_i) \neq \text{sign} \left(\left[\frac{1}{n} \sum_{j=1}^n S_G(\mathbf{v}_t^j) \right]_i \right) \right) \\
 &\leq \sum_{i=1}^d 2G \left| \frac{[\nabla f(\mathbf{x}_t)]_i}{G} - \left[\frac{1}{n} \sum_{j=1}^n S_G(\mathbf{v}_t^j) \right]_i \right| \\
 &= 2G \left\| \frac{\nabla f(\mathbf{x}_t)}{G} - \frac{1}{n} \sum_{j=1}^n S_G(\mathbf{v}_t^j) \right\|_1 \\
 &\leq 2G \sqrt{d} \left\| \frac{\nabla f(\mathbf{x}_t)}{G} - \frac{1}{n} \sum_{j=1}^n S_G(\mathbf{v}_t^j) \right\|.
 \end{aligned} \tag{13}$$

Rearranging and taking the expectation, we have:

$$\begin{aligned}
 \mathbb{E} \left[\left\| \frac{\nabla f(\mathbf{x}_t)}{G} - \frac{1}{n} \sum_{j=1}^n S_G(\mathbf{v}_t^j) \right\| \right] &\leq \mathbb{E} \left[\left\| \frac{\nabla f(\mathbf{x}_t)}{G} - \frac{1}{nG} \sum_{j=1}^n \mathbf{v}_t^j \right\| \right] + \mathbb{E} \left[\left\| \frac{1}{n} \sum_{j=1}^n \left(S_G(\mathbf{v}_t^j) - \frac{\mathbf{v}_t^j}{G} \right) \right\| \right] \\
 &\leq \mathbb{E} \left[\frac{1}{G} \left\| \nabla f(\mathbf{x}_t) - \frac{1}{n} \sum_{j=1}^n \mathbf{v}_t^j \right\| \right] + \sqrt{\mathbb{E} \left[\left\| \frac{1}{n} \sum_{j=1}^n \left(S_G(\mathbf{v}_t^j) - \frac{\mathbf{v}_t^j}{G} \right) \right\|^2 \right]} \\
 &\leq \mathbb{E} \left[\frac{1}{G} \left\| \nabla f(\mathbf{x}_t) - \frac{1}{n} \sum_{j=1}^n \mathbf{v}_t^j \right\| \right] + \sqrt{\frac{1}{n^2} \sum_{j=1}^n \mathbb{E} \left[\left\| S_G(\mathbf{v}_t^j) - \frac{\mathbf{v}_t^j}{G} \right\|^2 \right]} \quad (14) \\
 &\leq \mathbb{E} \left[\frac{1}{G} \left\| \nabla f(\mathbf{x}_t) - \frac{1}{n} \sum_{j=1}^n \mathbf{v}_t^j \right\| \right] + \sqrt{\frac{1}{n^2} \sum_{j=1}^n \mathbb{E} \left[\left\| S_G(\mathbf{v}_t^j) \right\|^2 \right]} \\
 &\leq \mathbb{E} \left[\frac{1}{G} \left\| \nabla f(\mathbf{x}_t) - \frac{1}{n} \sum_{j=1}^n \mathbf{v}_t^j \right\| \right] + \frac{\sqrt{d}}{\sqrt{n}},
 \end{aligned}$$

where the second inequality is due to the fact that $(\mathbb{E}[X])^2 \leq \mathbb{E}[X^2]$, and the forth inequality is because of $\mathbb{E}[S_G(\mathbf{v}_t^j)] = \mathbf{v}_t^j/G$, as well as the S_G operation in each node is independent. Rearranging the terms and summing up, we have:

$$\begin{aligned}
 \frac{1}{T} \sum_{i=1}^T \mathbb{E} [\|\nabla f(\mathbf{x}_t)\|_1] &\leq \frac{2(\Delta_f + L)}{\eta T} + 4\sqrt{d} \mathbb{E} \left[\frac{1}{T} \sum_{i=1}^T \left\| \nabla f(\mathbf{x}_t) - \frac{1}{n} \sum_{j=1}^n \mathbf{v}_t^j \right\| \right] + \frac{4dG}{\sqrt{n}} + 6\eta Ld \\
 &\leq \frac{2(\Delta_f + L)}{\eta T} + 4\sqrt{d} \sqrt{\mathbb{E} \left[\left\| \frac{1}{T} \sum_{i=1}^T \left\| \nabla f(\mathbf{x}_t) - \frac{1}{n} \sum_{j=1}^n \mathbf{v}_t^j \right\|^2 \right]} + \frac{4dG}{\sqrt{n}} + 6\eta Ld,
 \end{aligned}$$

where the last inequality is due to Jensen's Inequality. Note that in Theorem 4.5, we have already shown that

$$\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \left\| \frac{1}{n} \sum_{j=1}^n \mathbf{v}_t^j - \nabla f(\mathbf{x}_t) \right\|^2 \right] \leq \frac{2\sigma^2}{\beta_2 n T} + \frac{32\eta^2 L^2 d}{\beta_2^2} + \frac{2\beta_2 \sigma^2}{n}.$$

Finally, we can ensure that

$$\begin{aligned}
 \frac{1}{T} \sum_{i=1}^T \mathbb{E} [\|\nabla f(\mathbf{x}_t)\|_1] &\leq \frac{2(\Delta_f + L)}{\eta T} + \frac{4dG}{\sqrt{n}} + 6\eta Ld + 4\sqrt{d} \sqrt{\mathbb{E} \left[\left\| \frac{1}{T} \sum_{i=1}^T \left\| \nabla f(\mathbf{x}_t) - \frac{1}{n} \sum_{j=1}^n \mathbf{v}_t^j \right\|^2 \right]} \\
 &\leq \frac{2(\Delta_f + L)}{\eta T} + \frac{4dG}{\sqrt{n}} + 6\eta Ld + 4\sqrt{d} \sqrt{\frac{2\sigma^2}{n\beta_2 T} + \frac{2\sigma^2 \beta_2}{n} + \frac{32L^2 \eta^2 d}{\beta_2^2}}.
 \end{aligned}$$

By setting $\beta_2 = \frac{1}{2}$ and $\eta = \mathcal{O}(T^{-1/2}d^{-1/2})$, we have

$$\frac{1}{T} \sum_{i=1}^T \|\nabla f(\mathbf{x}_t)\|_1 = \mathcal{O} \left(\frac{(1 + \Delta_f + L)d^{1/2}}{T^{1/2}} + \frac{(1 + G + \sigma)d}{n^{1/2}} \right).$$

By setting $\beta_2 = \frac{1}{2}$ and $\eta = \mathcal{O}(n^{-1/2})$, we have

$$\frac{1}{T} \sum_{i=1}^T \|\nabla f(\mathbf{x}_t)\|_1 = \mathcal{O} \left(\frac{(1 + \Delta_f + L)n^{1/2}}{T} + \frac{(1 + G + \sigma + L)d}{n^{1/2}} \right).$$

G. Proof of Theorem 5.4

Since the function $f(\mathbf{x})$ is L -smooth, we have the following:

$$\begin{aligned}
 f(\mathbf{x}_{t+1}) &\leq f(\mathbf{x}_t) + \langle \nabla f(\mathbf{x}_t), \mathbf{x}_{t+1} - \mathbf{x}_t \rangle + \frac{L}{2} \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 \\
 &\leq f(\mathbf{x}_t) - \eta \left\langle \nabla f(\mathbf{x}_t), \mathbf{S}_1 \left(\frac{1}{n} \sum_{j=1}^n \mathbf{S}_G(\mathbf{v}_t^j) \right) \right\rangle - \eta \lambda \langle \nabla f(\mathbf{x}_t), \mathbf{x}_t \rangle + \frac{L}{2} \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 \\
 &= f(\mathbf{x}_t) + \eta \left\langle \nabla f(\mathbf{x}_t), \frac{\nabla f(\mathbf{x}_t)}{G} - \mathbf{S}_1 \left(\frac{1}{n} \sum_{j=1}^n \mathbf{S}_G(\mathbf{v}_t^j) \right) \right\rangle \\
 &\quad - \eta \left\langle \nabla f(\mathbf{x}_t), \frac{\nabla f(\mathbf{x}_t)}{G} \right\rangle + \frac{\eta G}{2} \lambda^2 \|\mathbf{x}_t\|^2 + \frac{\eta}{2G} \|\nabla f(\mathbf{x}_t)\|^2 + \frac{L}{2} \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 \\
 &= f(\mathbf{x}_t) + \eta \left\langle \nabla f(\mathbf{x}_t), \frac{\nabla f(\mathbf{x}_t)}{G} - \mathbf{S}_1 \left(\frac{1}{n} \sum_{j=1}^n \mathbf{S}_G(\mathbf{v}_t^j) \right) \right\rangle - \frac{\eta}{2G} \|\nabla f(\mathbf{x}_t)\|^2 + \frac{\eta G}{2} \lambda^2 \|\mathbf{x}_t\|^2 + \frac{L}{2} \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2.
 \end{aligned}$$

Taking expectations leads to:

$$\begin{aligned}
 &\mathbb{E}[f(\mathbf{x}_{t+1}) - f(\mathbf{x}_t)] \\
 &\leq \eta \mathbb{E} \left[\left\langle \nabla f(\mathbf{x}_t), \frac{1}{G} \nabla f(\mathbf{x}_t) - \mathbf{S}_1 \left(\frac{1}{n} \sum_{j=1}^n \mathbf{S}_G(\mathbf{v}_t^j) \right) \right\rangle \right] - \frac{\eta}{2G} \mathbb{E}[\|\nabla f(\mathbf{x}_t)\|^2] + \frac{\eta G}{2} \lambda^2 \|\mathbf{x}_t\|^2 + \frac{L}{2} \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 \\
 &= \eta \mathbb{E} \left[\left\langle \nabla f(\mathbf{x}_t), \frac{1}{G} \nabla f(\mathbf{x}_t) - \frac{1}{n} \sum_{j=1}^n \mathbf{S}_G(\mathbf{v}_t^j) \right\rangle \right] - \frac{\eta}{2G} \mathbb{E}[\|\nabla f(\mathbf{x}_t)\|^2] + \frac{\eta G}{2} \lambda^2 \|\mathbf{x}_t\|^2 + \frac{L}{2} \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 \\
 &= \eta \mathbb{E} \left[\left\langle \nabla f(\mathbf{x}_t), \frac{1}{G} \nabla f(\mathbf{x}_t) - \frac{1}{nG} \sum_{j=1}^n \mathbf{v}_t^j \right\rangle \right] - \frac{\eta}{2G} \mathbb{E}[\|\nabla f(\mathbf{x}_t)\|^2] + \frac{\eta G}{2} \lambda^2 \|\mathbf{x}_t\|^2 + \frac{L}{2} \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 \tag{15} \\
 &\leq \eta \mathbb{E} \left[\frac{1}{4G} \|\nabla f(\mathbf{x}_t)\|^2 + \frac{1}{G} \left\| \nabla f(\mathbf{x}_t) - \frac{1}{n} \sum_{j=1}^n \mathbf{v}_t^j \right\|^2 \right] - \frac{\eta}{2G} \mathbb{E}[\|\nabla f(\mathbf{x}_t)\|^2] + \frac{\eta G}{2} \lambda^2 \|\mathbf{x}_t\|^2 + \frac{L}{2} \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 \\
 &= \frac{\eta}{G} \mathbb{E} \left[\left\| \nabla f(\mathbf{x}_t) - \frac{1}{n} \sum_{j=1}^n \mathbf{v}_t^j \right\|^2 \right] - \frac{\eta}{4G} \mathbb{E}[\|\nabla f(\mathbf{x}_t)\|^2] + \frac{\eta G}{2} \lambda^2 \|\mathbf{x}_t\|^2 + \frac{L}{2} \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2.
 \end{aligned}$$

Additionally, we can ensure that

$$\begin{aligned}
 \|\mathbf{x}_t\| &= \|(1 - \eta\lambda)\mathbf{x}_{t-1} - \eta \text{sign}(\mathbf{v}_{t-1})\| \\
 &\leq (1 - \eta\lambda) \|\mathbf{x}_{t-1}\| + \eta \|\text{sign}(\mathbf{v}_{t-1})\| \\
 &\leq (1 - \eta\lambda) \|\mathbf{x}_{t-1}\| + \eta \sqrt{d} \\
 &\leq (1 - \eta\lambda)^{t-1} \|\mathbf{x}_1\| + \eta(t-1)\sqrt{d},
 \end{aligned}$$

By setting $\|\mathbf{x}_1\|_\infty \leq \eta T$, we also know that

$$\begin{aligned}
 \|\mathbf{x}_t\|^2 &\leq 2(1 - \eta\lambda)^{2(t-1)} \|\mathbf{x}_1\|^2 + 2\eta^2(t-1)^2 d \\
 &\leq 2 \|\mathbf{x}_1\|^2 + 2\eta^2(t-1)^2 d \\
 &\leq 2 \|\mathbf{x}_1\|_\infty^2 d + 2\eta^2(t-1)^2 d \\
 &\leq 2 \|\mathbf{x}_1\|_\infty^2 d + 2\eta^2 T^2 d \leq 4\eta^2 T^2 d.
 \end{aligned}$$

Rearranging the terms and summing up:

$$\begin{aligned}
 & \frac{1}{T} \sum_{i=1}^T \mathbb{E} [\|\nabla f(\mathbf{x}_t)\|^2] \\
 & \leq \frac{4\Delta_f G}{\eta T} + 4\mathbb{E} \left[\frac{1}{T} \sum_{i=1}^T \left\| \nabla f(\mathbf{x}_t) - \frac{1}{n} \sum_{j=1}^n \mathbf{v}_t^j \right\|^2 \right] + 2G^2 \lambda^2 \frac{1}{T} \sum_{i=1}^T \|\mathbf{x}_t\|^2 + \frac{2GL}{\eta} \frac{1}{T} \sum_{i=1}^T \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 \\
 & \leq \frac{4\Delta_f G}{\eta T} + 4\mathbb{E} \left[\frac{1}{T} \sum_{i=1}^T \left\| \nabla f(\mathbf{x}_t) - \frac{1}{n} \sum_{j=1}^n \mathbf{v}_t^j \right\|^2 \right] + 2G^2 \lambda^2 (4\eta^2 T^2 d) + \frac{2GL}{\eta} \left(6\eta^2 d + \frac{2\eta \lambda \|\mathbf{x}_1\|^2}{T} \right) \\
 & \leq \frac{4\Delta_f G}{\eta T} + 4\mathbb{E} \left[\frac{1}{T} \sum_{i=1}^T \left\| \nabla f(\mathbf{x}_t) - \frac{1}{n} \sum_{j=1}^n \mathbf{v}_t^j \right\|^2 \right] + 8G^2 \lambda^2 \eta^2 T^2 d + 12GL\eta d + \frac{4GL\lambda \|\mathbf{x}_1\|^2}{T} \\
 & \leq \frac{4(\Delta_f + L)G}{\eta T} + 4\mathbb{E} \left[\frac{1}{T} \sum_{i=1}^T \left\| \nabla f(\mathbf{x}_t) - \frac{1}{n} \sum_{j=1}^n \mathbf{v}_t^j \right\|^2 \right] + 20GL\eta d,
 \end{aligned}$$

where the last inequality is valid by setting $\lambda \leq \frac{\sqrt{L}}{T\sqrt{\eta G}}$ and $\lambda \eta \|\mathbf{x}_1\|^2 \leq 1$.

Note that in Theorem 4.5, we have already shown that

$$\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \left\| \frac{1}{n} \sum_{j=1}^n \mathbf{v}_t^j - \nabla f(\mathbf{x}_t) \right\|^2 \right] \leq \frac{2\sigma^2}{\beta_2 n T} + \frac{32\eta^2 L^2 d}{\beta_2^2} + \frac{2\beta_2 \sigma^2}{n}.$$

Finally, we obtain the final bound:

$$\begin{aligned}
 \mathbb{E} \left[\frac{1}{T} \sum_{i=1}^T \|\nabla f(\mathbf{x}_t)\| \right] & \leq \sqrt{\mathbb{E} \left[\frac{1}{T} \sum_{i=1}^T \|\nabla f(\mathbf{x}_t)\|^2 \right]} \\
 & \leq \sqrt{\frac{4(\Delta_f + L)G}{\eta T} + 20\eta L d G + \frac{8\sigma^2}{\beta_2 n T} + \frac{8\sigma^2 \beta_2}{n} + \frac{128L^2 \eta^2 d}{\beta_2^2}}.
 \end{aligned}$$

That is to say, by setting $\beta_2 = n^{1/3} \eta^{2/3} d^{1/3}$, $\eta = \mathcal{O} \left(\min \left\{ \frac{1}{T^{1/2} d^{1/2}}, \frac{n^{2/5}}{T^{3/5} d^{1/5}} \right\} \right)$, we can obtain the convergence rate of $\mathcal{O} \left(\sqrt{LG + L^2 + \sigma^2 + \Delta_f G} \cdot \max \left\{ \frac{d^{1/4}}{T^{1/4}}, \frac{d^{1/10}}{n^{1/5} T^{1/5}} \right\} \right)$.

H. Proof of Theorem 5.5

According to the previous analysis of Theorem 5.4, we have that

$$\frac{1}{T} \sum_{i=1}^T \mathbb{E} \|\nabla f(\mathbf{x}_t)\|^2 \leq \frac{4(\Delta_f + L)G}{\eta T} + 4\mathbb{E} \left[\frac{1}{T} \sum_{i=1}^T \left\| \nabla f(\mathbf{x}_t) - \frac{1}{n} \sum_{j=1}^n \mathbf{v}_t^j \right\|^2 \right] + 20\eta LdG.$$

Using the same analysis as in equation (9), we can ensure that

$$\mathbb{E} \left[\left\| \frac{1}{n} \sum_{j=1}^n \mathbf{v}_{t+1}^j - \nabla f(\mathbf{x}_{t+1}) \right\|^2 \right] \leq (1 - \beta_1) \mathbb{E} \left[\left\| \frac{1}{n} \sum_{j=1}^n \mathbf{m}_t^j - \nabla f(\mathbf{x}_t) \right\|^2 \right] + \frac{\beta_1^2 \sigma^2}{n} + \frac{2L^2}{\beta_1} \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2.$$

Also, in equation (11), we have already shown that

$$\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \left\| \frac{1}{n} \sum_{j=1}^n \mathbf{m}_t^j - \nabla f(\mathbf{x}_t) \right\|^2 \right] \leq \frac{\sigma^2}{n\beta_2 B_0 T} + \frac{2\sigma^2 \beta_2}{n} + \frac{16L^2 \eta^2 d}{n\beta_2},$$

As a result, by setting $\beta_2 \leq \beta_1 \leq \sqrt{\beta_2}$, we know that

$$\begin{aligned} \mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \left\| \frac{1}{n} \sum_{j=1}^n \mathbf{v}_t^j - \nabla f(\mathbf{x}_t) \right\|^2 \right] &\leq \frac{\sigma^2}{nT} + (1 - \beta_1) \mathbb{E} \left[\frac{1}{T} \sum_{t=1}^{T-1} \left\| \frac{1}{n} \sum_{j=1}^n \mathbf{m}_t^j - \nabla f(\mathbf{x}_t) \right\|^2 \right] + \frac{\beta_1^2 \sigma^2}{n} + \frac{2L^2}{\beta_1 T} \sum_{t=1}^T \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 \\ &\leq \frac{\sigma^2}{nT} + \frac{\sigma^2}{n\beta_2 B_0 T} + \frac{2\sigma^2 \beta_2}{n} + \frac{16L^2 \eta^2 d}{n\beta_2} + \frac{\beta_1^2 \sigma^2}{n} + \frac{12L^2 \eta^2 d}{\beta_1} \\ &\leq \frac{2\sigma^2}{\beta_2 n B_0 T} + \frac{32\eta^2 L^2 d}{\beta_2} + \frac{4\beta_2 \sigma^2}{n}, \end{aligned}$$

where $B_0 = 1$. Finally, we obtain the final bound:

$$\begin{aligned} \mathbb{E} \left[\frac{1}{T} \sum_{i=1}^T \|\nabla f(\mathbf{x}_t)\| \right] &\leq \sqrt{\mathbb{E} \left[\frac{1}{T} \sum_{i=1}^T \|\nabla f(\mathbf{x}_t)\|^2 \right]} \\ &\leq \sqrt{\frac{4(\Delta_f + L)G}{\eta T} + 20\eta LdG + \frac{2\sigma^2}{\beta_2 n B_0 T} + \frac{32\eta^2 L^2 d}{\beta_2} + \frac{4\beta_2 \sigma^2}{n}}. \end{aligned}$$

By setting $\beta_2 = \mathcal{O}\left(\frac{1}{T^{1/2}}\right)$, $\eta = \mathcal{O}\left(\frac{1}{T^{1/2}d^{1/2}}\right)$, we can obtain the convergence rate of $\mathcal{O}\left(\sqrt{LG + L^2 + \sigma^2 + \Delta_f G} \cdot \frac{d^{1/4}}{T^{1/4}}\right)$.

I. Additional Experiments

Additional Dataset We also evaluated our methods on the CIFAR-10 dataset using a ResNet-18 and ResNet-50 model. As shown in Table 4 and Figure 3, our Dis-Lion-VR algorithm consistently surpasses the other methods, achieving the highest overall accuracy.

Gradient Norm Analysis We plotted the gradient norm over iterations. As depicted in Figure 4, our proposed algorithms exhibit a smaller gradient norm, and Dis-SGD displays the largest.

Table 4. Accuracy on CIFAR-10 in the distributed setting.

Model	Optimizer	Accuracy(%)
ResNet-18	Dis-SGD	84.97 ± 0.04
	Dis-AdamW	85.36 ± 0.04
	Dis-Lion	85.36 ± 0.01
	Dis-Lion-VR	85.38 ± 0.01
	Dis-Lion-VR-CE	85.36 ± 0.02
ResNet-50	Dis-SGD	84.84 ± 0.06
	Dis-AdamW	86.00 ± 0.02
	Dis-Lion	86.15 ± 0.04
	Dis-Lion-VR	86.19 ± 0.02
	Dis-Lion-VR-CE	86.16 ± 0.01

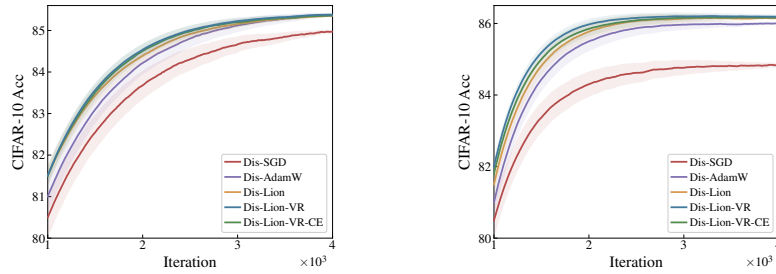


Figure 3. Accuracy in the Distributed Setting. CIFAR-10 accuracy curve of ResNet-18 (Left) and ResNet-50 (Right).

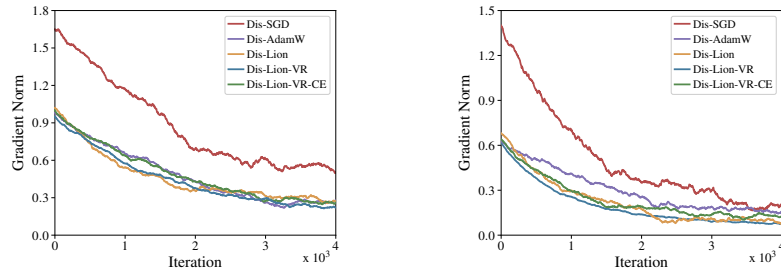


Figure 4. Gradient norm curve of ResNet-18 (Left) and ResNet-50 (right) on CIFAR-10.